Ariadne: Mapping the Way Back

A systems account of autonomous multi-agent collaborative visual mapping and 3D construction under unreliable communication links

Suvojit Das

OpenCV AI Competition 2026 submission

Code: https://github.com/CynicPixel/Ariadne Video: https://youtu.be/yrbcvQKnfkA

Implementation and evidence snapshot: 27 October 2026

Abstract

A major structural collapse can leave people trapped in spaces responders cannot safely enter. The "golden 72 hours" is an imperfect but familiar reminder that useful information has to arrive while it can still shape the search. In these spaces, a camera feed leaves important questions hanging: where is the map backed by observations, can a person fit through an opening, and will the apparent shortest way back stay clear?

Ariadne studies how a team of autonomous visual agents can decide what to observe, what to share, and which measurements to trust as they build a common 3D model of the space under a shared mission budget. The pipeline starts with image registration, feature tracking, and stereo-inertial pose estimation. Each agent builds a local map and pose graph; over constrained links, agents exchange compact place-recognition cues, then check candidate connections with plausible geometry and robust consensus before they can reshape the shared map. The team uses that map to pick informative views, allocate exploration tasks, and plan a responder-sized route based on estimated clearance and risk. An LLM orchestration harness helps manage task-level decisions, with human review available when an action crosses the approved risk envelope. Gaussian splat reconstruction lets incident command inspect the space from viewpoints the agents never visited.

Experiments show clear gains across key parts of this chain: learned matching improves pose estimation over an ORB baseline across the tested EuRoC flights; stereo-inertial mapping reaches the comparison class of established systems; robust collaboration resolves severe perceptual aliasing while preserving tested true map connections; and evaluated egress routes agree with ground-truth passability. Ariadne takes its name from the princess who gave Theseus the thread to trace his way through the Labyrinth. Its thread is made of observations, verified geometry, and a route that can lead responders in, and guide them home.

1 The Problem: Finding a Way Through the Labyrinth

After a collapse, the spaces that matter most may be the ones a human cannot safely enter: narrow voids, unstable floors, dark corridors, rooms cut off by rubble, and routes whose open-looking entrances hide an impassable pinch point. Responders need more than a camera feed. They need to know where the map is backed by observations, whether two agents are describing the same corridor, whether a person can fit through it, and whether the route back out is still viable.

Ariadne is for incident-command teams and search-and-rescue responders who need spatial decision support from small visual agents. The agents use stereo RGB, inertial measurements in the SLAM pipeline, and camera-mounted lighting. They build geometry and routes; identifying people and tracking their identities over time are outside the system's purpose.

The "golden 72 hours" gives the problem a familiar frame. It is a rule of thumb, not a promise or a stopwatch for every incident, but it captures the practical pressure: information has to arrive in time to affect a decision. A detailed map that turns up after the search has moved on is mostly an archival achievement.

The name Ariadne comes from the Greek myth. Princess Ariadne gives Theseus a thread to find his way through the Labyrinth and back out after facing the Minotaur. Ariadne aims to give responders a thread of its own: a route grounded in what the agents have observed, updated as the picture changes, and clear about where the evidence runs thin.

The research question is practical: when sensing, movement, communication, energy, and mission time are all limited, which observation should the team acquire next, which relationship should it transmit, and which geometric evidence should be allowed to move the shared map? Our hypothesis was that uncertainty- and connectivity-aware coordination would outperform local coverage maximization under the same mission costs.

Ariadne follows that question from the camera to the route home. Image restoration and feature tracking feed stereo-inertial pose estimation; keyframes and landmarks form local maps, refined by bundle adjustment and pose-graph optimization. Agents exchange compact retrieval cues over limited links, prioritize candidate connections under a communication budget, and use geometric consensus to stop false matches from bending the shared map. That joint model helps the team choose informative views and allocate tasks. A volumetric map estimates clearance and risk, and a responder-scale planner searches for a viable way out. An orchestration harness helps choose among proposed tasks, with human review available when an action crosses the risk gate. Gaussian splats make the explored space easier to inspect.

The thread starts with perception and ends with a route. For a responder to follow it, the connections along the way have to hold.

2 What was built

The work began with a thin end-to-end build: a repeatable simulated mission that could produce camera and IMU data, estimate motion, record the run, and replay the same evidence. ROS 2 Jazzy and Gazebo Harmonic provide the mission environment; MCAP carries recorded sensor streams, trajectories, and scene ground truth into repeatable evaluation. This first thread made interfaces concrete early. Images, poses, maps, collaboration messages, planning outputs, and reconstruction chunks cross explicit data boundaries, so each layer can be exercised from a recorded run before it is joined to the next.

From that base, the visual path grew into a keyframe-based stereo-inertial estimator. It tracks features between frames, triangulates local landmarks, refines recent poses and points with windowed bundle adjustment, and connects keyframes in a pose graph. The estimator and graph optimizer are Ariadne implementations built from OpenCV 5 and Eigen primitives. EuRoC [16] and TUM sequences supply public-data checks; GTSAM serves as an independent test oracle for selected graph problems, not as a runtime dependency. The next section follows the front end from the initial ORB/LK build to the adopted tracking and place-recognition components, and the section after it details local stereo-inertial estimation.

The next layer let agents build partial maps separately, then decide which cross-agent evidence was worth communicating and which candidate joins could safely influence the shared graph. Experiments shaped the final choices: retrieval and matching are selective, messages are budgeted, and geometric checks reject inconsistent joins before global optimization. On the resulting map, a volumetric layer estimates observed support, clearance, and a structural-risk proxy. Candidate views are scored for exploration and assigned to agents; a separate responder-scale search finds a route to an exit through known traversable space. The communication, planning, and egress sections describe those layers and their measured limits in turn.

Two services sit above this deterministic core. An LLM harness can choose among planner proposals and issue constrained directives through an MCP-facing service; it does not estimate geometry or execute unchecked commands. A risk gate is designed to route qualifying actions to human review, while validation and fallback checks remain in ordinary code. Gaussian submaps provide an inspectable visual representation alongside the geometric map. Their live transport and browser display are demonstrated separately from GPU training, which was evaluated on a synthetic corridor. The later sections keep those evidence boundaries visible: the diagram summarizes the system's connections, while each subsystem section states what its own experiments establish and what has not yet been demonstrated in one integrated mission.

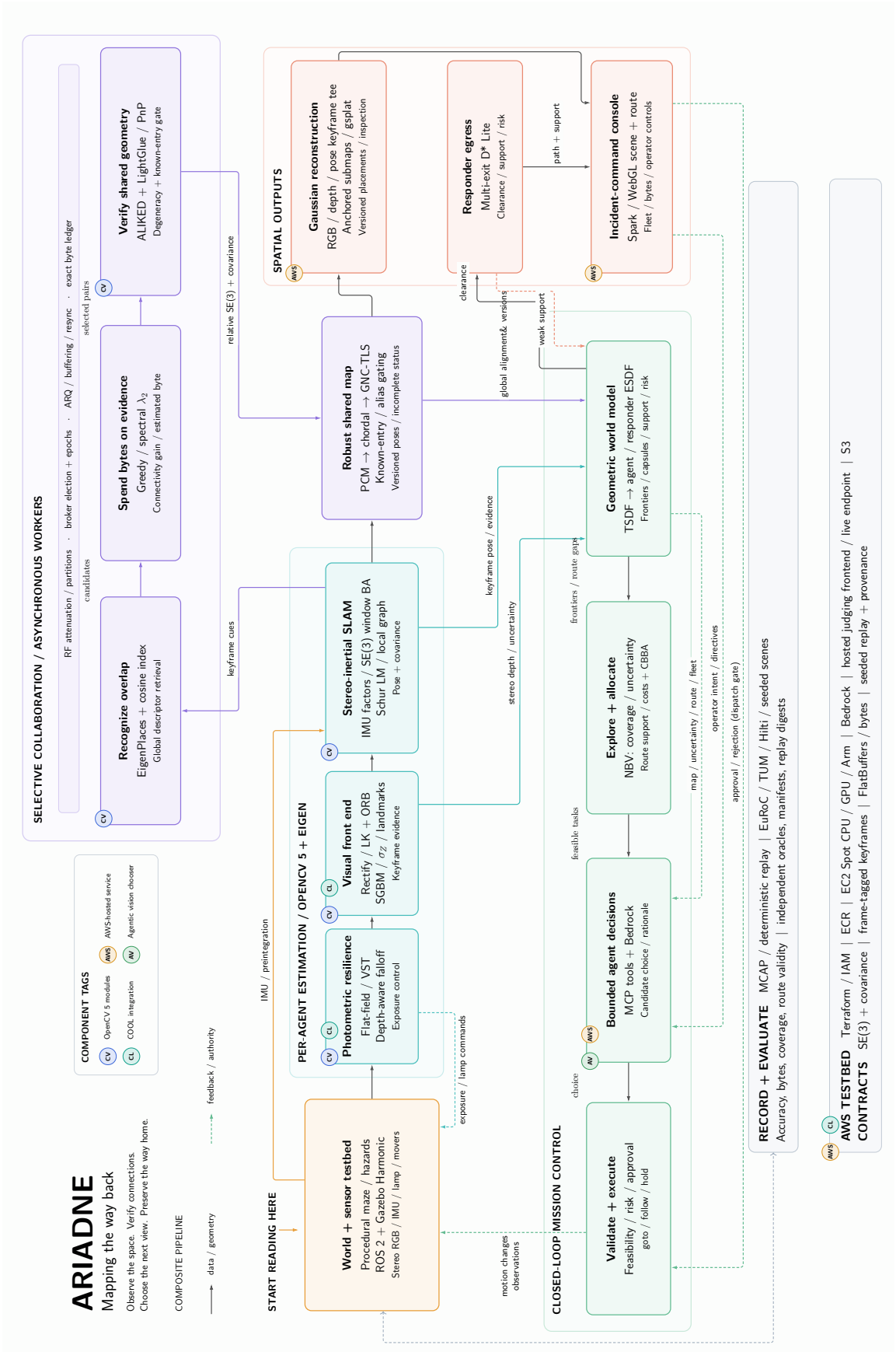


Figure 1: Ariadne's system architecture, from stereo-inertial sensing and OpenCV 5 estimation through selective collaboration, volumetric mapping, mission orchestration, responder egress, and Gaussian reconstruction. Solid arrows carry data and geometry; dashed arrows denote feedback, control, or replay. The lower rails identify evaluation and AWS services.

3 Visual front end and place recognition

Ariadne separates continuous visual tracking from the more expensive task of recognizing a previously visited place. Pyramidal Lucas–Kanade (LK) carries features between frames, while ORB replenishes tracks and supports landmark recovery. At keyframes, EigenPlaces retrieves candidate overlaps; ALIKED and LightGlue establish correspondences for selected pairs, which undergo geometric verification before contributing constraints to the shared map. This division concentrates learned inference where a successful recognition can correct accumulated drift or connect two agents' maps.

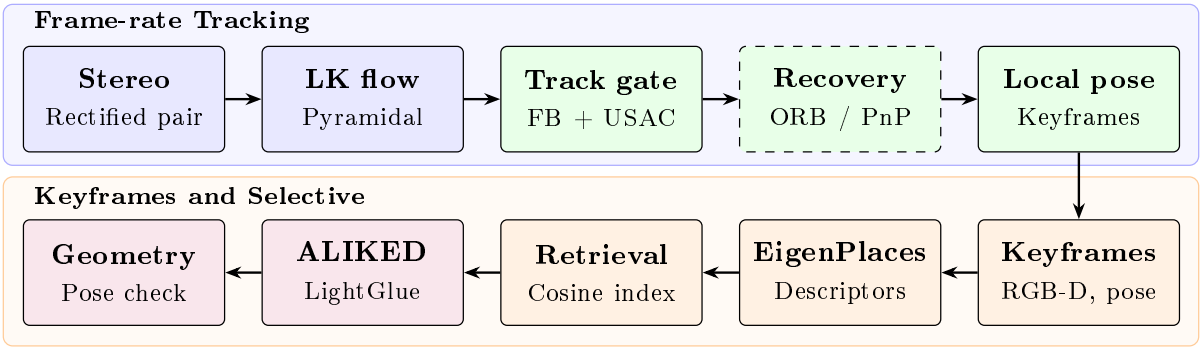


Figure 2: Ariadne tracks and validates features on every frame, replenishing ORB tracks and re-associating landmarks with PnP when needed. At keyframes, EigenPlaces retrieves candidate overlaps; ALIKED–LightGlue and geometric verification run only on selected pairs.

The front end evolved through measurement. The initial ORB/LK pipeline was followed by an OpenCV DNN implementation of XFeat, assembled from upstream weights and checked against reference outputs. XFeat reduced tracking losses but increased trajectory error on six of seven previously passing EuRoC flights. Changes to keyframe cadence, descriptor matching, and landmark re-association did not eliminate the regression. The adopted EuRoC visual-odometry configuration therefore retains ORB, reserving learned features for place recognition and selective pairwise matching [6, 7, 8, 9, 10].

Each frame passes through three-level pyramidal LK with a 21-pixel window, forward/backward consistency checks, and seeded USAC fundamental-matrix estimation to reject epipolar outliers. An 8×6 spatial grid distributes up to 1,000 ORB keypoints, with 400 active tracks as the replenishment target. When landmark-bearing tracks fall below 30, recovery matches current observations against healthy, untracked landmarks in the local window. Mutual nearest-neighbour matching, a 0.75 ratio test, and a Hamming-distance cap of 50 precede seeded PnP verification requiring at least 12 inliers. Recovered tracks consequently retain an established 3D observation history.

At each keyframe, EigenPlaces produces a unit-normalized, 256-dimensional descriptor. An exact cosine index retrieves candidate overlaps, which the communication policy can prioritize by expected graph-connectivity gain and transmission cost. Exact and approximate search returned identical top-ranked places on all 297 evaluated EuRoC queries; exact search required only microseconds for thousands of descriptors. Selected pairs proceed to ALIKED-n16, capped at 1,024 keypoints at native image resolution, and the nine-layer LightGlue matcher. The exported models reproduce upstream keypoints within 10^{-4} pixels and yield identical LightGlue match sets. Increasing the keypoint cap to 2,048 brought no measured pose gain at approximately 2.8× the cost.

> **Verification path.** LK tracking and ORB recovery maintain local landmark continuity. EigenPlaces proposes revisits; ALIKED–LightGlue supplies correspondences; seeded geometric estimation, PnP, degeneracy checks, and robust graph consistency determine which proposed connections can influence the map.

OpenCV 5 supplies the computational primitives: `cv::buildOpticalFlowPyramid` and `cv::calcOpticalFlowPyrLK` for tracking; `cv::ORB`, `cv::ALIKED`, and `cv::LightGlueMatcher` for features and matching; `cv::dnn::Net` for EigenPlaces and the XFeat experiment; and seeded USAC through `cv::findFundamentalMat` and PnP. Ariadne implements track management, landmark re-association, descriptor indexing, inference wrappers, and geometric acceptance policy. The pinned OpenCV 5.0.0 build lacks a native `cv::XFeat` class; its network was assembled over OpenCV DNN primitives [11].

On 4,815 EigenPlaces-retrieved EuRoC keyframe pairs across 11 flights, including 676 ground-truth-positive pairs, ALIKED–LightGlue increased pose AUC at 10° from 0.284 for ORB to 0.607, matching or exceeding

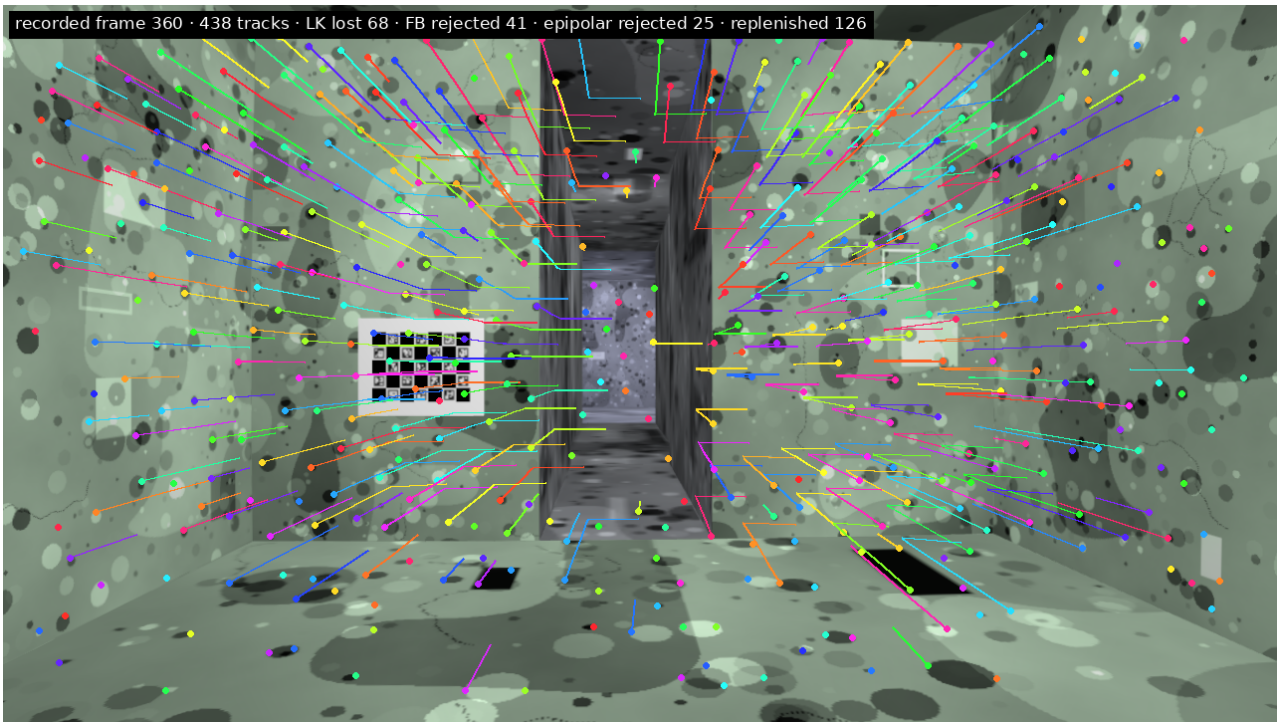


Figure 3: The front end on a recorded agent-0 camera frame (lit maze, recorded run `c51ab2`). Each colour is one track; the trail joins its positions over the previous sampled frames. Tracks were replayed offline with OpenCV 5 `calcOpticalFlowPyrLK` on every 20th recorded frame (window 21, four pyramid levels, forward–backward threshold 0.5 px, `findFundamentalMat` RANSAC at 1 px), and the banner shows that frame's track statistics. Script: `figures/src/make\_stills.py`.

ORB on every flight. Geometric average precision, scored against the fixed set of positives, rose from 0.303 **to** 0.807; LightGlue contributed another 0.042 AUC over ALIKED with mutual-nearest-neighbour matching. Four confidently incorrect essential-matrix verifications nevertheless survived with 210–264 inliers and rotation errors of 37°–59°. Their failure establishes the importance of the downstream PnP, degeneracy, and consistency gates, particularly in visually repetitive interiors.

For place retrieval, EigenPlaces achieved 0.626 **R@1** at 376×240, against 0.444 for mean-bit descriptors and 0.167 for random retrieval. On 61 revisit queries from one clean simulator scene, its 640×360 export reached 0.705 R@1, with no observed top-1 benefit from full resolution. Four-thread CPU measurements gave p50 inference times of 311.8 ms for EigenPlaces at 640×360, 1,998.6 ms for ALIKED at 1280×720, and 921.0 ms for a warm LightGlue pair. These costs justify the event-driven learned path: LK sustains frame-to-frame estimation, while expensive recognition is invoked when its evidence can materially change the map. ORB remains the matched pairwise baseline here; the complete estimator is compared with published stereo-inertial systems in the following section.

4 Stereo-inertial estimation and local SLAM

Estimator architecture

Ariadne combines stereo geometry with inertial motion estimates in a sliding-window visual-inertial estimator. Stereo supplies metric depth; IMU preintegration carries rotation, velocity, and position information between keyframes. The estimator maintains a local landmark map and emits pose and covariance estimates to a local SE(3) pose graph.

Stereo geometry and landmark management

Ariadne matches rectified stereo images, estimates disparity with Semi-Global Block Matching, and triangulates landmarks in metric coordinates. Landmark admission checks parallax, depth, reprojection error, stereo uncertainty, and relative position uncertainty. Tracks that survive across keyframes add multi-view observations. After optimization, high-residual observations are removed; landmarks are culled according to whether

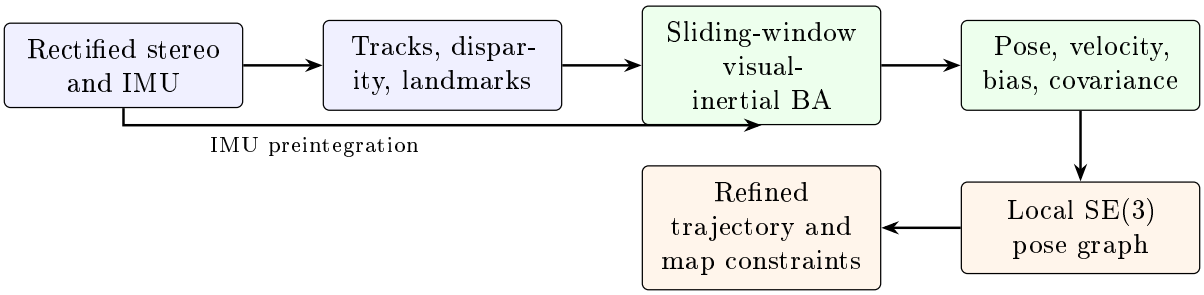


Figure 4: Stereo and inertial measurements constrain a local sliding window; its pose and covariance outputs form odometry edges in the local pose graph.

they were observed when predicted visible.

Online row refinement compensates for residual vertical misalignment between cameras. On run V2_01, we reduced the median stereo row residual from **1.46 px to 0.09 px**. Left-right exposure matching also helps preserve correspondences when the cameras' automatic exposure differs.

Sliding-window visual-inertial optimization

The adopted window contains ten keyframes. Each camera pose is represented on SE(3); full inertial mode adds world-frame velocity, gyro bias, and accelerometer bias. Visual residuals are reprojection errors in both rectified cameras. Consecutive keyframes receive preintegrated IMU factors and bias random-walk factors. Visual residuals use uncertainty weighting and a Huber loss.

A keyframe score combines translation, rotation, parallax, track loss, and pose uncertainty, with hard limits on track count and elapsed time. The oldest keyframe anchors the window gauge. Analytic Jacobians assemble the damped normal equations. Ariadne Schur-eliminates the independent 3D landmark blocks, solves the reduced pose system, and back-substitutes landmark updates. Selected covariance blocks are recovered through linear solves rather than an explicit inverse. When a keyframe leaves the window, its relative pose and transported covariance become an odometry edge for the local pose graph.

Initialization and recovery

IMU samples are preintegrated between keyframes using the on-manifold formulation of Forster et al. [1]. Each inertial edge summarizes rotation, velocity, and position change, including propagated uncertainty and bias sensitivity. Initialization follows ten consecutive measured keyframes: Ariadne estimates gyro bias from visual and inertial rotations, solves for velocity, gravity, and accelerometer bias, then refines gravity magnitude.

Gravity is estimated at initialization and held fixed in the world frame while the window refines velocity and biases. We tested estimating gravity tilt independently in each window, but short-window tilt and accelerometer bias were weakly distinguishable; allowing both to drift made the trajectory less stable. The adopted solution keeps the initialized gravity direction fixed.

When tracking is lost, Ariadne records pose provenance and uses inertial and visual bridging to support recovery. Gyro-only rotation while lost was an earlier estimator mode; full preintegration in the window is the adopted EuRoC configuration.

Implementation and validation

OpenCV 5 supplies stereo disparity, triangulation, PnP, and the Levenberg–Marquardt driver. Ariadne implements the inertial model, residuals, analytic Jacobians, Schur backend, covariance handling, and local graph state. Eigen provides the reduced-system and graph factorizations.

Numerical checks agreed with independent references: the pose-graph solver matched GTSAM 4.3 [4] on **100/100** seeded synthetic graphs, with worst relative energy difference 1.5×10^{-12}. A one-free-pose bundle-adjustment case matched OpenCV's PnP refinement within 1.3×10^{-9} **m** and 2.8×10^{-9} **rad**. Estimator outputs were bit-identical across OpenCV thread-pool settings of 2, 5, and 16 threads for three trajectories of 4,001 poses each.

Estimator loop. Initialize velocity, gravity, and biases from ten measured keyframes. For each frame, track stereo features and validate landmarks. When the keyframe policy fires, add the pose, landmarks, and preintegrated IMU factor; run up to ten damped LM iterations with landmark Schur elimination;

Operation	OpenCV 5 connection	Ariadne implementation
Stereo depth and geometry	`StereoSGBM`, `triangulatePoints`, `solvePnPRansac`	Landmark admission and uncertainty policy
Nonlinear optimization	`cv::detail::LevMarqBase`	Visual-inertial factors, analytic Jacobians, Schur elimination
Pose refinement checks	`solvePnPRefineLM`, `projectPoints`	Jacobian and solver cross-checks
IMU fusion	No OpenCV inertial primitive	Preintegration, initialization, bias and gravity estimation

Table 1: OpenCV provides geometry and the optimization driver; Ariadne supplies the visual-inertial estimation problem.

recover selected pose covariance; and emit a relative-pose odometry edge. During tracking loss, retain pose provenance and use the configured visual and gyro bridging.

EuRoC evaluation

The loop-closed and VIO-only rows below come from the **same 11-flight run**, with the same seed, keyframe-refined ATE, and alignment procedure. Ariadne's VIO-only row reproduces the earlier run to four decimal places. ATE is computed over the refined trajectory after SE(3) alignment, without scale adjustment. The loop-closed system improves mean ATE from **0.120 m to 0.066 m**.

EuRoC flight	Ariadne VIO	Ariadne + loops	ORB-SLAM3	Class ceiling
MH_01	0.068	**0.021**	0.036	0.166
MH_02	0.044	**0.028**	0.033	0.152
MH_03	0.099	**0.033**	0.035	0.125
MH_04	0.223	**0.046**	0.051	0.280
MH_05	0.140	**0.044**	0.082	0.284
V1_01	0.054	0.038	0.038	0.076
V1_02	0.047	**0.027**	0.014	0.110
V1_03	0.329	0.288	0.024	0.120
V2_01	0.039	**0.030**	0.032	0.070
V2_02	0.110	0.071	0.014	0.100
V2_03	0.169	0.097	0.024	0.190
Mean	**0.120**	**0.066**	**0.035**	**0.138**

Table 2: EuRoC RMS ATE in metres. The per-flight class ceiling is the worst published result among ORB-SLAM3, VINS-Fusion, Kimera, and BASALT; the aggregate ceiling is 0.138 m.

By that explicit comparison rule, Ariadne VIO is within the published class on **9/11** flights, and the loop-closed system on **10/11**. The loop-closed system matches or improves on ORB-SLAM3's published per-flight result on **7/11** flights. Its mean remains above ORB-SLAM3's published 0.035 m mean due to the outlier run V1_03 skewing the mean.

This is a published-baseline comparison, not a same-binary rerun: Ariadne's figures are a single seed-1 run, while reference numbers are taken from the systems' published results. ORB-SLAM3 is a full SLAM system, and BASALT's reported mean covers ten flights; these differences matter when interpreting the table. The visual-inertial formulation, comparison systems, and dataset are described in the original papers [16, 12, 1, 2, 3].

5 Communication-aware collaborative mapping

A shared map is only as reliable as the connections between its local maps. Finding those connections begins with recognizing overlapping places, continues with deciding which evidence is worth transmitting, and ends with determining which geometric constraints can safely reshape the shared graph. In collapsed interiors, repeated corridors and similar openings make false connections especially costly: one plausible but incorrect loop

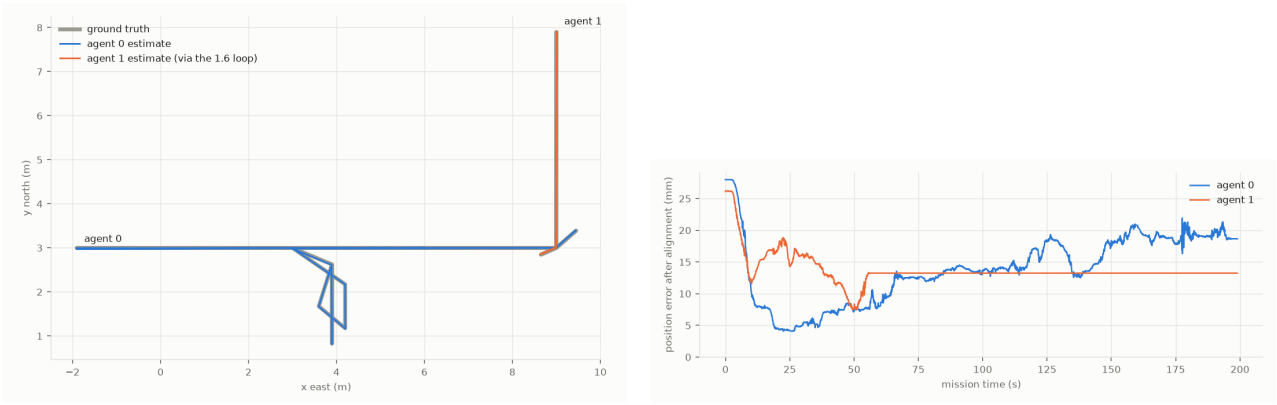


Figure 5: Two-agent simulated mission from the AWS pre-merge gate (c7i spot, run `20261024T083919Z-cpux86-73e3fa`, 54/54 checks). Left: estimated trajectories over ground truth, with agent 1 placed through the inter-agent loop. Right: position error after alignment over the 200 s mission, in millimetres.

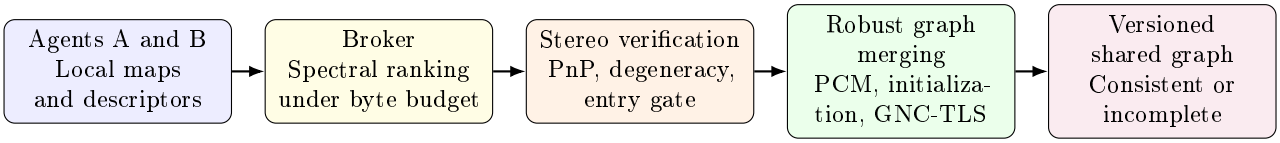


Figure 6: Ariadne's collaborative mapping path. Compact descriptors identify candidate overlaps; budgeted feature exchanges provide geometric evidence; robust estimation determines which connections enter the shared graph. The evaluated configuration uses a brokered solve with explicit incomplete states during communication loss.

closure can displace an entire agent trajectory. Ariadne couples communication selection with robust estimation, accounting for serialized bytes, rejecting inconsistent geometry, and preserving an explicitly incomplete map when connectivity is lost.

Architecture and communication-aware selection

Each agent maintains an independent stereo-inertial map and publishes compact EigenPlaces descriptors when links permit. Candidate overlaps are ranked before the broker requests ALIKED–LightGlue feature evidence. Feature payloads are cached by keyframe, allowing several candidate edges to reuse one transmission. This two-stage exchange reserves expensive correspondence data for potentially useful connections and makes communication cost part of the mapping decision.

Ariadne evaluates candidate edges through their expected improvement to pose-graph algebraic connectivity, $\lambda_2(L)$, the second-smallest eigenvalue of the graph Laplacian. For an edge $e = (i, j)$ with expected weight w_e, its first-order spectral gain is

$$\Delta\lambda_2(e) \approx w_e(v_i - v_j)^2, \qquad s(e \mid S) = \frac{\Delta\lambda_2(e)}{c(e \mid S)}, \tag{1}$$

where v is the Fiedler vector and $c(e \mid S)$ is the marginal wire cost given already selected exchanges S. The broker greedily selects high-scoring candidates subject to the byte budget, with local one-swap refinement. A simpler greedy policy and exact enumeration on small instances provide comparison points. This follows the task-oriented exchange principle of Tian et al. [17], applied here to graph connectivity and measured serialized traffic.

Geometric verification and robust graph merging

A retrieved match must first survive stereo relative-pose estimation and PnP verification. Ariadne examines pose covariance, translation and rotation marginals, parallax, and the spatial distribution of supporting 3D points. Poorly constrained hypotheses, including nearly collinear correspondences, are rejected before they can influence graph consistency.

For missions with known relative agent starts, the entry briefing provides an additional geometric constraint.

Each candidate's visual transform is compared with the transform predicted from the entry prior and accumulated agent odometry. Covariances are propagated and transported into a common tangent frame, producing a six-dimensional innovation gate:

$$r_e = \mathrm{Log}\Big((\hat{T}_e^{\mathrm{prior}})^{-1} T_e^{\mathrm{visual}}\Big), \qquad r_e^\top \Sigma_e^{-1} r_e \le \chi^2_6(0.999). \tag{2}$$

Here Σ_e includes the relevant prior, odometric, and visual uncertainties after propagation. The gate exploits information available at deployment and is enabled only for the evaluated known-start condition. Unknown relative starts remain vulnerable to perceptual aliasing.

Surviving hypotheses enter pairwise consistency maximization (PCM). Two closures are compatible when their composition with the intervening odometry-only paths yields a sufficiently small cycle residual. A consistency graph records these relationships, and a deterministic greedy clique heuristic selects mutually supported constraints; exact clique search is available for small instances. PCM removes isolated false closures efficiently, although a self-consistent cluster of incorrect matches can survive [18].

The selected constraints initialize a shared $SE(3)$ pose graph. Without an entry prior, sparse chordal least squares estimates rotations, projection onto $SO(3)$ restores valid rotation matrices, and a translation Laplacian supplies the remaining initialization. With known starts, the entry transforms anchor the agent frames before joint refinement. Ariadne then applies graduated non-convexity with truncated least squares (GNC-TLS), progressively adjusting loop-edge weights according to their whitened residuals while retaining quadratic odometry factors. This layered backend draws on PCM, robust pose-graph estimation, and the GNC formulation [20, 21].

> **Collaborative update.** Broadcast compact descriptors; retrieve overlaps and prioritize candidate exchanges by spectral gain per marginal byte. Request cached feature payloads, verify stereo geometry and PnP, and reject degenerate hypotheses. Apply the known-entry Mahalanobis gate when its prior is available. Construct pairwise consistency relationships, select PCM-compatible closures, initialize the joint graph, and refine with GNC-TLS. Publish the resulting graph with an epoch, version, and completeness state.

Asynchronous communication and implementation

A deterministic priority rule elects a broker within each connected neighborhood. Epochs distinguish broker membership across partitions; store-and-forward queues preserve messages during disconnection, while acknowledgements and selective fragment retransmission protect graph data. On reconnection, agents reconcile graph snapshots under the current epoch. Missing keyframes or edges leave the shared graph marked incomplete until their dependencies arrive. This sparse exchange architecture follows the collaborative principles of Swarm-SLAM, with brokered optimization and versioned recovery in Ariadne [19].

Wire costs are recorded at the serialization boundary, including framing and compression. Cached feature payloads and precomputed odometry compositions reduce repeated matching and PCM work; agents exchange compact graph state instead of image histories. OpenCV 5 supplies stereo geometry, learned matching, PnP, and rotation-matrix primitives. Ariadne implements the protocol, epoch handling, byte ledger, spectral selection, PCM, chordal initialization, and GNC-TLS backend, using Eigen for sparse spectral and linear-algebra operations.

Experimental results

The most consequential result concerns false map merges. In scene 0, the ungated configuration produced worst seed-mean merged ATE of 3.78–3.79 m across four policy–budget cells. Enabling the known-entry gate reduced this to 0.014675 **m**; the worst individual gated run reached 0.0212 m. Across both scenes, the gate retained all 560/560 **tested true edges**. Ten additional scene-0 true edges were never tested by the gate and are excluded from this retention result. The improvement reflects the combined gated configuration, including its effect on graph initialization.

Communication selection was evaluated independently on 60 small synthetic instances with exact optima. Spectral selection achieved mean selected-to-optimum utility of 0.970, against 0.936 for greedy; worst-case observed ratios were 0.650 and 0.533. These finite-instance results establish the quality of the implemented heuristic without implying an approximation guarantee. In the gated simulator sweep, spectral selection matched or improved greedy in three of four cells, with ATE differences below approximately 1 mm; the fourth cell slightly favored greedy.

In scene 1, budgeted exchange consumed 2.27–2.45 MB of first-transmission traffic and yielded 6.8–8.6 mm merged ATE across three seeds. The request-everything similarity configuration consumed 14.8–16.0 MB; one

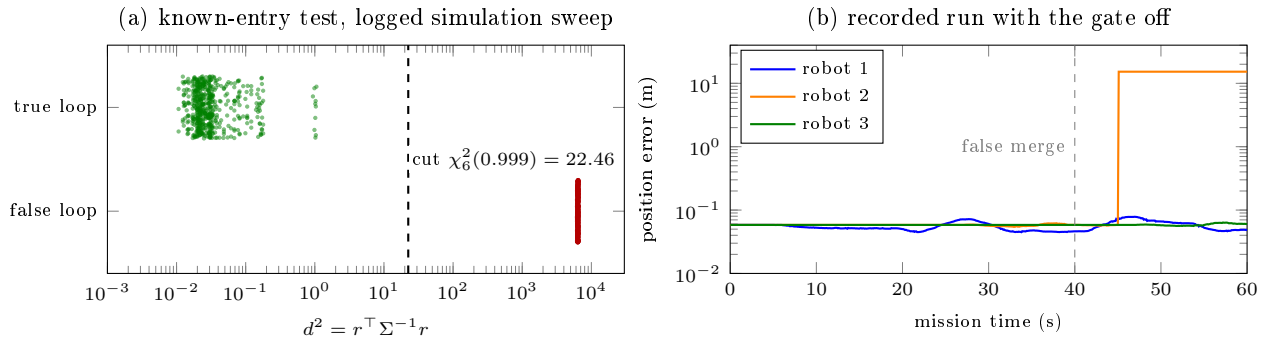


Figure 7: Why the known-entry gate exists. (a) Every geometry-verified inter-agent candidate in the logged sweep (`evaluation/runs/collab\_d178/logged\_4ec3f6`): 560 true loops (pose error $\leq 10°$) score $d^2 \leq 1.1$, 1,164 false loops score $d^2 \geq 6{,}175$, against a 22.46 cut. (b) A recorded three-robot run with the gate off (`collab60d`): 32 of 491 candidates pass every geometric check, all 32 are false, and a false merge at $t = 40\,\text{s}$ leaves robot 2 15.3 m off. In a second recorded run with the gate on, it refused 8 of 8 tested candidates and every robot stayed within 7.5 cm of ground truth. Data: `figures/data/*.csv`, derived by `figures/src/make\_stills.py`.

Aspect	Ariadne evidence	Relationship to prior work
Budgeted exchange	0.970 mean spectral utility ratio against exact optima on 60 synthetic instances.	Tian et al. establish task-oriented exchange; Ariadne evaluates Laplacian connectivity against measured serialization costs [17].
Robust merging	560/560 tested true edges retained by the known-entry gate; scene-0 worst seed-mean ATE reduced to 1.47 cm.	PCM and Kimera-Multi motivate consistency-based rejection and robust optimization. Ariadne adds a deployment-specific entry prior, evaluated without a common-dataset comparison [18, 20].
Communication protocol	Broker epochs, selective retransmission, store-and-forward, and explicit incomplete graph states.	Swarm-SLAM motivates sparse collaborative exchange; Ariadne uses a versioned brokered solve with replayable partition recovery [19].

Table 3: Architectural and algorithmic comparison. Reported Ariadne measurements come from its own synthetic and simulated evaluations; the cited systems use different experimental settings.

seed suffered 3.09 m aliasing error, while two runs failed to report a complete multi-agent merge. The configurations also differ in entry-gate use, so their separation in accuracy cannot be attributed to communication selection alone.

Byte prediction proved less reliable than the serializer ledger. All 24 budgeted request rows met the 15% prediction-error threshold, with a worst error of 10.65%. Response estimates exceeded that threshold in 8/24 rows, leaving the overall class-total accounting gate unmet. Actual transmitted bytes remain measured independently of those estimates.

Controlled robust-backend tests retained all ten planted true inter-agent loops and rejected every injected false loop across five seeds. PCM followed by GNC-TLS survived the tested outlier ramp through 90% false loops in groups of ten, and through 70% with two growing false clusters. GNC alone also handled the tested ramp at approximately ten times the runtime. The differing breakpoints expose the dependence of PCM on the structure of mutually consistent false hypotheses.

Comparison with prior systems

Together, these experiments identify the condition under which sparse collaboration becomes dependable. Communication-aware selection substantially reduces transmitted evidence, while geometric verification, the entry prior, PCM, and GNC-TLS limit the damage caused by ambiguous visual matches. Known-start scenes demonstrate a reliable shared graph at low communication cost; unknown-start aliasing and imperfect byte prediction remain measured limits of the approach.

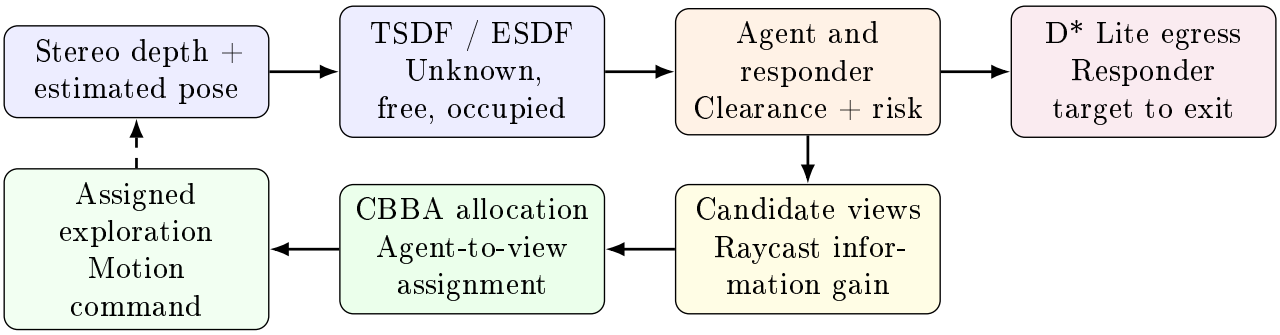


Figure 8: The map-to-action loop. Exploration assigns informative views to agents and incorporates their subsequent observations; responder-scale egress planning branches from the same volumetric model.

6 Exploration, mapping, and task allocation

Ariadne closes the loop between observing a space and deciding where to look next. Stereo depth and estimated poses accumulate into a volumetric map; uncertainty, clearance, and connectivity then guide candidate views and multi-agent task allocation. The same geometry supports a second decision: whether a responder can reach a validated exit from a location of interest. Exploration seeks useful observations under mission constraints, while egress planning depends on observed, body-sized passages.

Volumetric map and compute path

Stereo observations enter a 5 cm voxel-hash truncated signed distance field (TSDF), organized into 8^3-voxel blocks. Each voxel stores a signed distance D_v and accumulated confidence weight W_v, updated by uncertainty-weighted ray integration within a three-voxel truncation band. A zero weight preserves the unknown state, distinguishing unobserved space from measured free space. The map maintains separate Euclidean signed distance fields (ESDFs): agent planning treats occupied cells as obstacles, while responder planning additionally excludes unknown cells. Exact one-dimensional squared-distance transforms along each voxel axis recover 3D Euclidean clearance, with incremental recomputation around changed regions. A 0.5 m coarse layer and maintained frontier clusters support longer-range exploration.

Resolution follows the geometry of the intended user. At 8 and 10 cm, one-cell wall bleed closed a 1.05 m entrance to a 0.40 m-radius responder in both tested ground-truth scenes; 5 cm preserved the passage. Against Ariadne's earlier OctoMap-based placeholder on the same evidence spool, TSDF coverage improved from 0.933 to 0.942, free-space precision from 0.961 to 0.979, and occupied-space precision from 0.707 to 0.865. Mean per-keyframe integration cost was 16.1 ms; incremental ESDF and frontier updates averaged 64.3 and 48.3 ms on the development host.

The block-hash fusion and frontier-update designs draw on Voxblox and FUEL [23, 24]. Ariadne implements voxel storage, weighted fusion, distance queries, and incremental updates because the required per-voxel access is absent from the evaluated OpenCV volume interface, while OpenCV's distance transform operates in 2D. `cv::PCA` partitions large frontier clusters, `cv::filter2D` computes local support statistics, and `cv::parallel_for\_` distributes candidate scoring. The earlier occupancy baseline uses an OctoMap-style representation [22].

Body-aware traversability and structural-risk proxy

Responder clearance requires a different interpretation of unknown geometry. A strict observed-volume test would reject much of a standing body above a camera operating near 1 m height. Ariadne combines a 2.5D ground lattice with 3D capsule collision checks, requiring observed ground support, sufficient planar clearance, and unobstructed body geometry. The evaluated responder has radius 0.40 m and height 1.80 m; agents use their own capsule and altitude constraints.

Evaluation covers every ground-truth-labelled cell of the aligned ground grid, including 12,334 traversable cells. Unknown predictions count against recall, and any claimed passage through a ground-truth obstacle counts against precision. The hybrid test achieved 1.000 **precision and** 0.900 **recall**, with no false traversability claims and valid connections from both tested starts to the target. Strict 3D observed-volume clearance achieved the same precision but only 0.354 recall, closing both routes; unobserved voxels caused 91% of its missed traversable cells. The earlier placeholder reached 0.898 recall. These results describe the evaluated

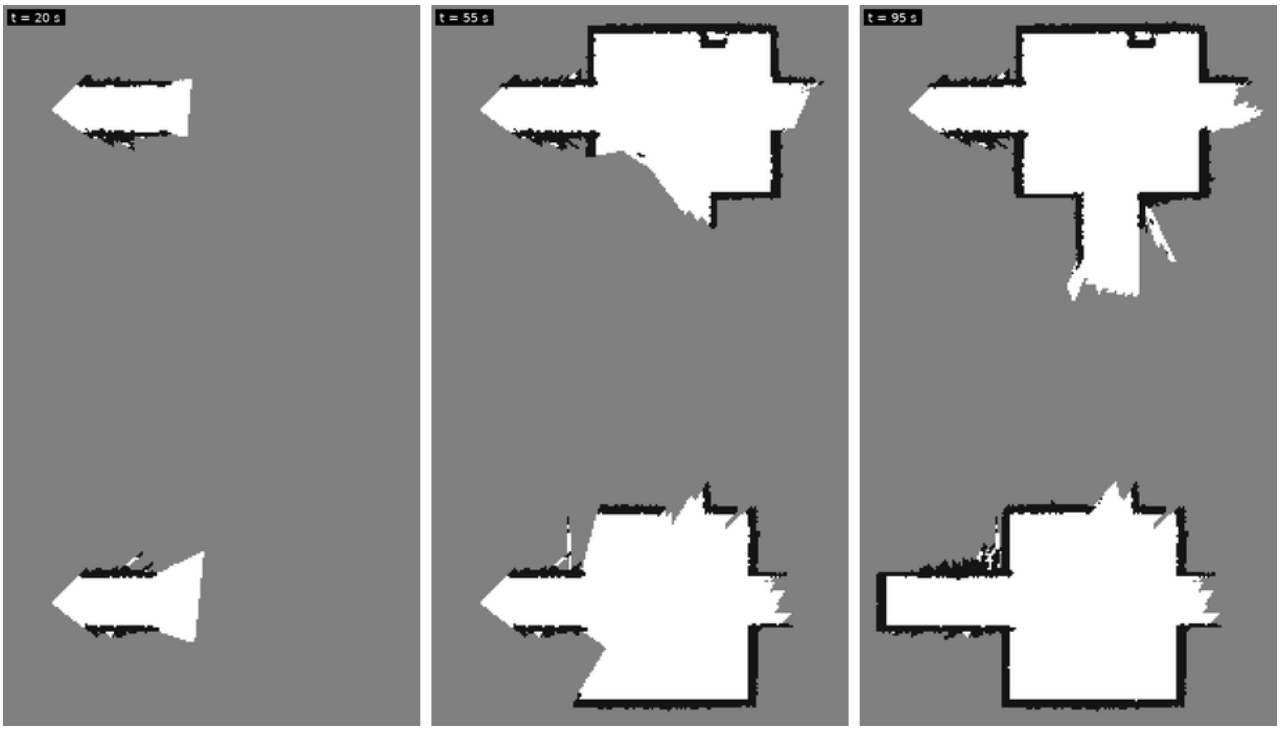


Figure 9: The volumetric map growing during the final-stack recorded run (two agents, `maze\_short\_6`, 200 s, AWS). Occupancy slab of the Phase-4 TSDF/ESDF map at three times: white is observed free space, black is surface, grey is unknown.

ground-plane grid and its observation geometry.

A separate structural-risk proxy estimates overhanging occupied mass and poorly supported surfaces above an otherwise open walking column:

$$r = 1 - (1 - r_{\text{overhang}})(1 - r_{\text{unsupported}}). \tag{3}$$

Against procedural hazard labels, the combined proxy achieved ROC AUC 0.831, exceeding the preregistered 0.75 threshold. Its operating threshold nevertheless saturated at 1.0, producing a 0.338 false-positive rate in traversable cells outside labelled zones. The overhang term alone reached 0.922 AUC with 0.156 false-positive rate, but remains an unelected alternative.

Risk enters the route objective, so calibration affects the route itself. In one hazard seed, risk-weighted planning avoided the labelled zone at unchanged path length (4.991 m), reducing estimated route risk from 0.340 to zero. In another, an already safe route lengthened by 4.4% without reducing estimated risk. The proxy therefore supplies a measurable geometric warning signal, with false positives that can impose unnecessary detours. It does not establish structural integrity or predict collapse.

Next-best-view planning

Each map update can expose new frontiers, weakly observed coarse cells, unrevisited peer keyframes, suspected place aliases, or poorly supported egress segments. These five sources generate candidate observations. A cylindrical sampler places viewpoints at radii 1.0, 1.5, and 2.0 m, across 16 azimuths at 1.0 m height. Collision and line-of-sight tests remove infeasible views; two viewpoints per target advance to detailed scoring.

Ariadne estimates expected information gain by raycasting candidate camera frusta into the volumetric map. A 64-ray pass ranks views by coverage, followed by 256 rays over a 2.84 m depth horizon for the retained candidates. Voxels are deduplicated within each frustum. The utility combines expected coverage, entropy reduction, egress support, graph-connectivity improvement, and alias disambiguation:

$$G(a) = \widehat{\Delta C}(a) + \widehat{\Delta H}(a) + \widehat{\Delta E}(a) + \widehat{\Delta P}(a) + \widehat{\Delta L}(a), \tag{4}$$

$$U(a) = G(a) - \lambda_{\text{path}}(t) C_{\text{path}}(a) - C_{\text{bandwidth}}(a) - C_{\text{risk}}(a). \tag{5}$$

Each gain term is normalized by its current candidate-set 90th percentile, falling back to its maximum when necessary; an identically zero term contributes zero. Observation-weight entropy rewards poorly constrained surfaces even after their first sighting. Terms lacking a producer in the evaluated configuration remain zero. The path-cost weight increases with mission time and energy expenditure, while a hard reserve rule restricts proposals to return or hold whenever remaining energy falls below 1.3 times the estimated safe-return requirement. Independent targets are evaluated in parallel.

The preregistered coverage-versus-time hypothesis failed. Across 36 closed-loop runs spanning two scenes, six policies, and three seeds, Ariadne ranked third by coverage AUC in both scenes: 0.884 on `ci\_tiny` and 0.440 on `maze\_short`, against 0.892/0.495 for coverage-greedy and 0.895/0.443 for the TARE-inspired policy. Ariadne eventually led all six policies on `ci\_tiny` after 80 s and attained the highest `maze\_short` coverage at 200 s (0.757, versus 0.713 and 0.681). Its slower initial progress nevertheless reduced area under the coverage-time curve.

Execution traces help explain the gap. Agents spent 45–46% of mission time rotating and 31–32% translating. Approximately half the turning arose from heading reversals between approaching a viewpoint and facing its observation target. An arrival-heading option and time-aware command cost have been implemented, alongside fixes for at-cell views and cut commands; the time-aware variant has yet to complete the full evaluation. The measured comparison therefore retains the original path-cost policy.

The TARE-inspired comparator schedules coarse 6 m subspaces and constructs local tours through nearest-neighbour insertion and 2-opt refinement [25]. On 200 synthetic tours of at most eight nodes, its worst score was 0.811 of exact enumeration. Its simulator results characterize this adapted policy rather than the published TARE system.

Multi-agent task allocation

Candidate views become stable tasks shared between agents. Ariadne uses event-triggered Consensus-Based Bundle Algorithm (CBBA) with bundle length $L_t = 1$. Each agent computes the value of an ordered task bundle p using a time-discounted, diminishing-gain score:

$$S_i(p) = \sum_{j \in p} \bar{c}_{ij} \exp\left(-\frac{\lambda_{\text{path}} d_{ij}(p)}{P_{\text{path}}} \right), \tag{6}$$

where $\bar{c}_{ij}$ incorporates task information value after bandwidth and risk costs, $d_{ij}(p)$ is geodesic travel distance within the ordered bundle, and P_{path} is a shared scale frozen for each auction event. Agents exchange winner, bid, and timestamp state when tasks appear, complete, or communication topology changes. Sparse consensus messages carry only tasks with assigned winners. A broker-side sequential greedy allocator (SGA) uses identical scores and tie-breaking as a reference. The score is designed for CBBA's diminishing-marginal-gain assumptions; its published convergence conditions additionally depend on communication and consensus behavior [15].

Across 40 two-agent replay replans, CBBA and SGA agreed on every assignment. For the 12 linked replans permitting exact enumeration, CBBA reached mean 0.994 of optimum, with worst ratio 0.983. Sparse consensus messages averaged 226 bytes and consumed 11.8 kB overall, compared with 123.4 kB for dense exchange. Partition behavior was exercised in synthetic tests; the replay establishes agreement and byte savings under its evaluated two-agent connectivity conditions.

7 Safe egress planning and replanning

Ariadne's egress planner turns the shared map into a route a responder could follow home. Given a location of interest and validated exits, it searches for a connected passage with sufficient body clearance and observational support. Routes occupy a 2D ground lattice, with 3D collision checks against the responder ESDF. Unknown or blocked columns are excluded from the graph, so uncertainty penalties can favor better-observed passages without admitting unverified ones.

Route graph and cost

The lattice is 8-connected, with diagonal moves constrained by corner occupancy. Each node represents a responder-traversable column, and each legal edge $x \to y$ carries a cost

$$c(x,y) = \|x-y\|\Bigg[1 + w_u(1-s_{xy}) + w_r r_{xy} + w_c \frac{\max(0, \rho_{\text{safe}} - d_{xy})}{\rho_{\text{safe}}}\Bigg], \tag{7}$$

where s_{xy} is observed support over the responder-body envelope, r_{xy} is the structural-risk proxy, and d_{xy} is ESDF clearance. Nonnegative penalties ensure $c(x,y) \geq \|x-y\|$, preserving the consistency of the octile-distance heuristic. Cost weights were fixed before evaluation. The main egress spool used $w_r = 0$; risk-weighted routing was evaluated separately after the proxy passed its preregistered calibration threshold.

Route confidence is the fifth percentile of support along the selected path, exposing short, poorly observed stretches that a mean would conceal. It summarizes geometric observation support; calibration against real traversal outcomes remains outside the evaluation.

Incremental multi-exit search

Ariadne uses multi-source D* Lite, propagating cost-to-exit backward from every traversable exit. Exit nodes have rhs = 0; other nodes maintain the one-step Bellman estimate

$$\text{rhs}(u) = \min_{v \in \text{Succ}(u)} [c(u,v) + g(v)], \qquad \text{key}(u) = \begin{pmatrix} \min(g, \text{rhs}) + h(u,s) \\ \min(g, \text{rhs}) \end{pmatrix}, \tag{8}$$

where s is the fixed responder target and h is octile distance. Only locally inconsistent vertices, $g(u) \neq \text{rhs}(u)$, enter the priority queue. Once the target is consistent, the route follows successive legal neighbors minimizing $c(u,v) + g(v)$.

Changes to occupancy, clearance, or support invalidate affected edges and trigger repairs to their incident vertices. Ariadne updates each changed cell's 3×3 neighborhood because diagonal corner legality also depends on adjacent orthogonal cells. The search reuses unaffected cost-to-go estimates, with no moving-start correction for the fixed target [26]. Tests exercise multiple exits, blocked corners, and doorway closure after route construction.

Evaluation and replanning

Across 180 randomized occupancy and edge-cost updates, incremental D* Lite reproduced fresh multi-source Dijkstra solutions exactly. It expanded 17,564 **vertices**, against 132,932 for repeated fresh searches, a 7.6× reduction.

On the recorded mission spool, all 12/12 **routes** were ground-truth traversable. The final 4.44142 m route agreed with the 4.4414215 m ground-truth geodesic, and the first valid route appeared at 145 s. Incremental search nevertheless averaged 0.46 ms per query against 0.04 ms for fresh A* on this short, nearly straight problem. End-to-end replanning averaged 545 ms (p95 651 ms), including support and traversability updates. The available profile does not isolate their individual costs.

The mission spool contained no obstruction discovered directly on an active route. A separate doorway-closure test verified path replacement or explicit no-path reporting. Thus, the mission establishes route validity and repeated map-to-route operation; the controlled test establishes response to route invalidation.

Risk-aware routing exposed the cost of imperfect calibration. In one hazard seed, the planner avoided the labelled hazard without increasing its 4.991 m path length, reducing estimated risk from 0.340 to zero. In another, saturation of the risk proxy lengthened an already hazard-free route by 4.4%, with no reduction in estimated risk. Geometric risk can therefore improve route choice, but false positives can also impose unnecessary detours.

Visibility-graph alternative

A FAR-inspired dynamic visibility graph was evaluated as a lightweight alternative. On 12 short-spool queries, its string-pulled route matched the lattice result at 4.403 m; graphs contained at most 31 nodes, with mean update and query times of 0.7 and 0.01 ms. A later hazard case exposed a missed passage: the visibility graph returned 11.115 m against a 4.424 m lattice route. This counterexample prevents a general path-quality claim for the implemented graph.

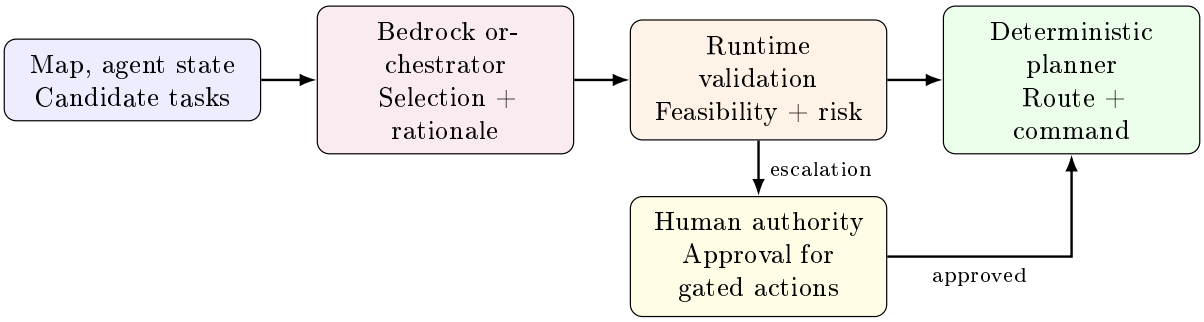


Figure 10: Task-level orchestration and intended authority boundary. The model selects planner-generated candidates; runtime validation and, where required, human approval precede deterministic dispatch. The separate egress planner computes responder routes directly from the geometric map.

FAR motivates efficient visibility-graph maintenance and permits exploratory routes through unknown regions [27]. Ariadne retains the conservative lattice for reported responder egress, where every route segment must satisfy known-space traversability.

OpenCV 5 boundary

The volumetric clearance and incremental search algorithms are Ariadne implementations. OpenCV 5 supplies adjacent geometric operations: `cv::filter2D` for support and risk aggregation, `cv::findContours` and `cv::approxPolyDP` for experimental visibility-graph construction, and the upstream stereo-depth primitives. TSDF fusion, 3D ESDF maintenance, route costs, and D* Lite remain explicit components of the planning system.

8 Language-model task decisions and human authority

Ariadne's language-model layer enters where the geometric planner has already identified useful actions. At each replan, the deterministic mission layer constructs feasible observation tasks from the current map, agent poses, uncertainty, and resource estimates. A multimodal model examines this bounded decision context and selects among the proposals. Its choices pass through runtime validation before becoming motion commands, keeping geometric feasibility, collision checks, and responder egress within the established planning system.

Decision architecture and authority

The orchestrator receives one joint decision request per replan, containing candidate tasks, agent assignments, route summaries, uncertainty, and a compact visual map. It returns candidate indices with a short rationale and may propose directives to prioritize or exclude regions, assign tasks, or hold agents. Directives require operator confirmation and take effect at a subsequent planning boundary. Candidate validation checks membership, feasibility, risk, and assignment uniqueness before dispatch.

Exploration decisions return to the sensing loop as movement produces new observations and changes the next candidate set. Responder egress remains an independent map query, allowing a validated exit route to be computed regardless of the model's exploration choice.

Runtime, tools, and approval

The orchestration service exposes image, map, uncertainty, route, and candidate queries through the Model Context Protocol (MCP). Amazon Bedrock Converse invokes Nova 2 Lite for multimodal selection; a configured DeepSeek text model provides a provider-error fallback. A browser-based demonstration console presents live map and agent state, planning decisions, and operator controls, while the service layer manages model calls and tool responses. Request and response payloads, tool calls, candidate selections, directives, and state digests are recorded for deterministic replay. Replay reproduces the recorded decisions; fresh model inference remains stochastic.

Actions requiring review are identified by a heuristic combining clearance and estimated structural risk:

$$
\begin{aligned}
r &= \max(r_{\text{geom}}, r_{\text{structural}}), \\
r_{\text{geom}} &= \text{clamp}\left(\frac{\rho_{\text{responder}} - c_{\min}}{\rho_{\text{responder}} - \rho_{\text{agent}}}, 0, 1\right).
\end{aligned}
\tag{9}
$$

Here $c_{\min}$ is estimated clearance and $r_{\text{structural}}$ is the geometric risk proxy. The approval threshold, fixed at 0.5, is a policy parameter on this normalized score. The formula requires finite clearance and $\rho_{\text{responder}} > \rho_{\text{agent}} > 0$. Startup configuration does not yet enforce this domain, leaving a validation defect in the gate.

The model-choice validator, approval request, and rejection paths are implemented and exercised in scripted and Bedrock-stub tests. Live runs remained below the threshold, leaving operator approval, rejection, and timeout unexercised with a real model. Code review also exposed a dispatch defect: after rejection, the original joint allocation can remain scheduled despite a fallback label. The computed risk-filtered alternative is not consistently substituted and revalidated. Consequently, the intended approval boundary is implemented only partially, and the live evidence does not establish rejection-safe dispatch. Duplicate assignments are checked within each joint decision; execution across retries and failures has no exactly-once guarantee.

Live evaluation and reproducibility

An initial two-agent, 90-second mission in `maze\_short\_6` completed 18 joint replans and 36/36 **Nova 2 Lite assignments**, without model errors, fallbacks, or runtime crashes. Its two recorded replays passed all 15 checks covering inputs, estimator state, directives, movement, and trace hashes. The initial artifact carried a dirty-tree and configuration-provenance inconsistency, motivating a matched comparison from a common source revision.

The subsequent classical–Bedrock pair fixed scene, seed, renderer, runtime, and mission configuration at revision `c143db14`, changing only the chooser. Both arms completed 18 joint replans and 36 agent decisions, with passing replay verdicts. Bedrock differed from the classical agent-0 choice on 6/18 replans and selected rank-zero candidates 31/36 times, compared with 26/36 for the classical arm. Each arm issued 36 commands, of which 34 were accepted and two remained unanswered at termination; none were refused.

A longer matched pair extended the same experiment to 500 simulated seconds. Both arms completed 100 joint replans, issued 200 commands, and recorded 10,001 frames per agent without unpaired-frame losses. Two replays per arm reproduced the live state digests and matched each other exactly. These experiments establish sustained live orchestration, behavioral differences between choosers, and reproducible recorded execution. Ground-truth coverage, egress validity, and mission completion were not scored between arms, leaving comparative mission benefit unresolved.

Model inference also carries a measurable operational cost. Across the 18 decisions of the matched Bedrock run, orchestration latency had median 5.8 s, p90 7.45 s, and maximum 9.03 s, excluding C++ planning and motion. The trace recorded 2,936,424 input tokens over 118 Bedrock calls. Tool exchanges repeatedly resubmit context, so this cumulative count exceeds the unique decision input; billing reconciliation was not performed. The matched 90-second missions required 611 s and 616 s of wall time for the classical and Bedrock arms, respectively. The 500-second recordings are indexed under `evaluation/runs/agentic\_planning\_pair\_500s\_20261025/`.

Relationship to language-guided navigation

Language-guided exploration provides useful architectural precedents. L3MVN and LFG score navigation frontiers using language-model reasoning; SayNav grounds task plans in scene graphs and delegates motion to conventional navigation; Co-NavGPT extends vision-language task assignment to cooperating robots [34, 35, 36, 37]. Ariadne applies bounded model selection to candidates generated from geometric uncertainty, connectivity, communication cost, and risk. Its matched experiments demonstrate that language-model decisions can alter the exploration schedule while remaining replayable through the deterministic runtime. Establishing whether those changes improve completed missions requires ground-truth outcomes across additional scenes and seeds, together with a verified approval and fallback path.

9 Gaussian-splat reconstruction: an inspectable surface model

Ariadne's Gaussian map gives incident command a view of the explored space beyond the agents' recorded camera positions. Each Gaussian encodes a 3D mean, anisotropic scale, orientation, opacity, and colour; differentiable rasterization projects these primitives into novel views. The representation complements the

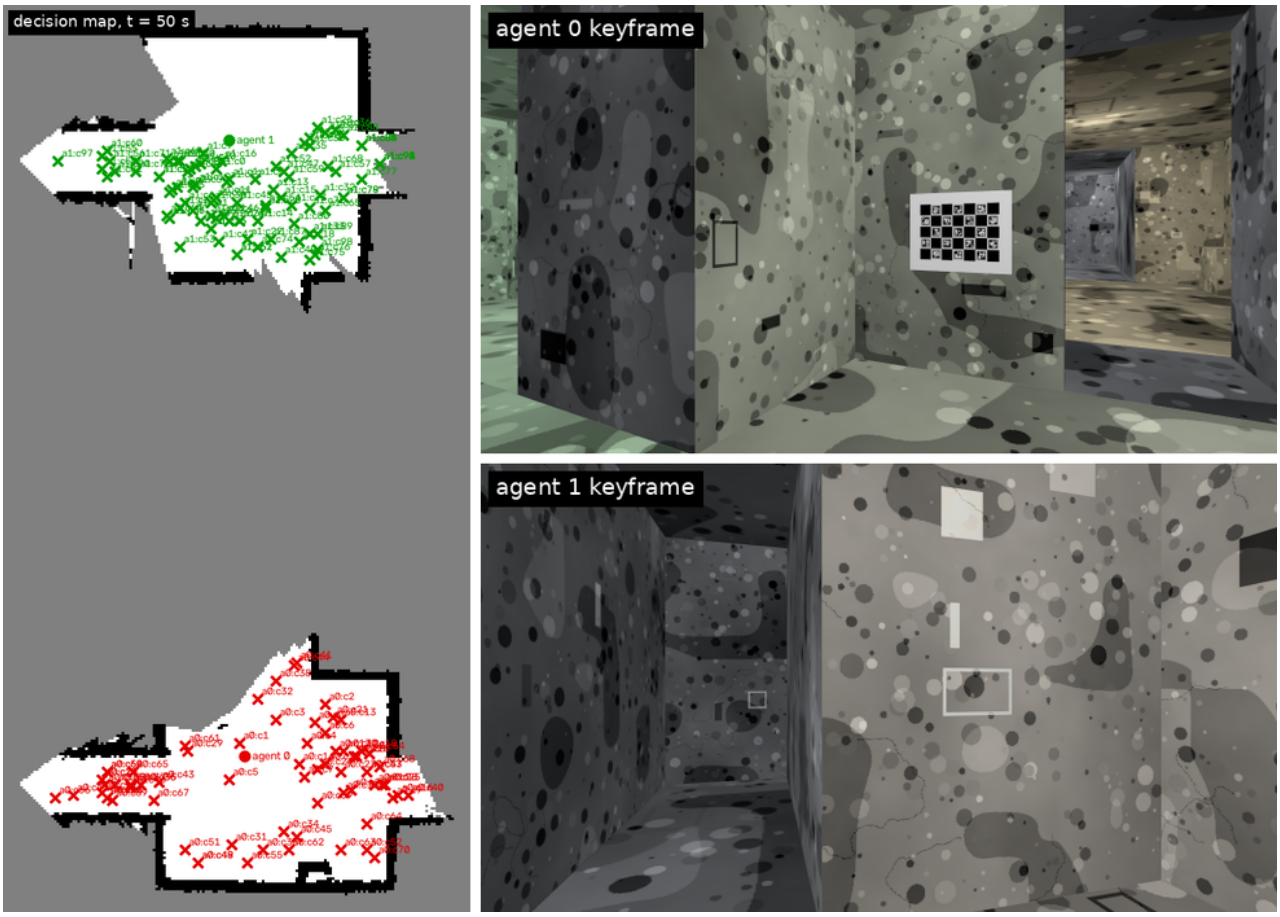


Figure 11: What the orchestrator sees at one decision of the final-stack recorded run ($t = 50\,\mathrm{s}$): the OpenCV-rendered map returned by `observe\_image`, with each agent's candidate goals, and the two agents' keyframes at that instant. That run made 40 joint replans and 80 Nova 2 Lite decisions, all validated by the C++ checks, with a passing replay.

geometric map used for clearance and egress. Reconstruction consumes validated poses and stereo depths, allowing the visual scene to evolve independently of localization and route planning.

The service receives rectified RGB, metric depth, depth uncertainty, and keyframe poses. It initializes Gaussians from stereo observations, groups them into anchor-relative submaps, optionally refines active submaps on a GPU, and exports compact chunks to a Spark/WebGL browser viewer. Training and browser rendering have separate compute budgets.

Submap frames and live transport

Each agent maintains submaps in the coordinate frame of an anchor keyframe. For agent odometry frame O_k, submap anchor A_s, and keyframe camera C, a depth sample is initialized through

$$\begin{aligned} T_{A_sC} &= T_{O_kA_s}^{-1} T_{O_kC}, \quad X_{A_s} = T_{A_sC} X_C, \\ T_{WA_s} &= T_{WO_k} T_{O_kA_s}. \end{aligned} \tag{10}$$

Gaussians remain in anchor coordinates. The browser applies T_{WA_s} once per submap, allowing an updated global placement to move the entire chunk without rewriting its Gaussian parameters. Submaps close after 30 keyframes or at correction boundaries. Rigid re-anchoring preserves internal Gaussian geometry; corrections within a submap require further refinement. This bounded-submap design follows the geometric separation explored by LoopSplat [31].

A nonblocking runtime tee sends length-prefixed `#kf` records over a local socket, with an eight-keyframe queue per agent and accounted oldest-first drops. Records contain agent and keyframe IDs, timestamps, pose versions, camera geometry, rectified RGB, and half-resolution float16 depth Z and axial uncertainty $S = \sigma_Z$. Placement records carry T_{WO_k}. The uncertainty follows stereo propagation,

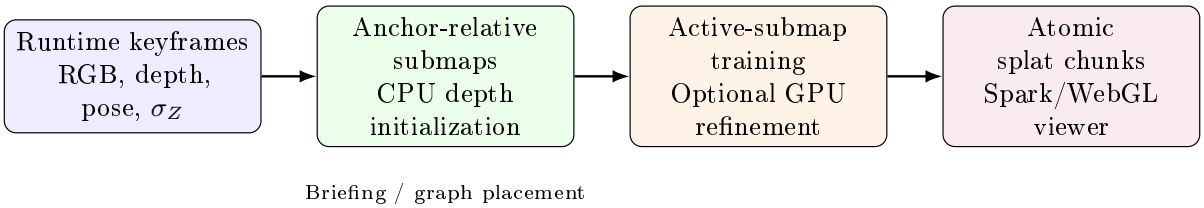


Figure 12: Reconstruction from validated stereo keyframes to browser-visible Gaussian submaps. The live demonstration exercised CPU initialization and streaming; GPU refinement was evaluated separately.

$$\sigma_Z = \frac{Z^2}{f_x B}\sigma_d, \tag{11}$$

where f_x, B, and σ_d denote focal length, baseline, and disparity-error proxy. Pose uncertainty is excluded from this estimate. The receiver validates payloads, frame conventions, and agent placement before accepting keyframes.

The service atomically publishes a placement table and one `.splat` chunk per submap. Each Gaussian occupies 32 bytes with degree-zero spherical harmonics; the table carries chunk identity, anchor, pose version, Gaussian count, and world transform. Active chunks may be replaced, while closed chunks remain fixed. Spark 2.3.1 renders one mesh per submap, applying placement independently of chunk contents.

Depth seeding and Gaussian refinement

The initial fused-cloud renderer established the reconstruction interface but produced soft, incomplete surfaces. Subsequent experiments adopted a Gaussian-SLAM-style depth-seeding and joint appearance–geometry objective [29]. The first keyframe samples valid depth uniformly and from high-gradient image regions, using fractions 100000/307200 and 50000/307200, respectively. Later keyframes target pixels whose rendered alpha is below 0.6. Samples require $0 < Z \leq 6\,\mathrm{m}$ and $\sigma_Z \leq 0.05\,\mathrm{m}$; a nearest-neighbour check excludes points within 1 cm of existing active-submap Gaussians. Accepted points receive local-distance scales, source-image colours, and initial opacity 0.5.

The active submap is optimized with gsplat 1.5.3 and PyTorch. Its adopted objective is

$$\mathcal{L} = 0.8\mathcal{L}_{1,\mathrm{colour}} + 0.2\mathcal{L}_{\mathrm{DSSIM}} + \mathcal{L}_{1,\mathrm{depth}} + \mathcal{L}_{\mathrm{isotropy}}, \tag{12}$$

where $\mathcal{L}_{\mathrm{DSSIM}} = 1 - \mathrm{SSIM}$ and the isotropy regularizer penalizes deviations of each Gaussian scale axis from its mean. Colour and metric-depth residuals share a finite-value, uncertainty-thresholded pixel mask; depth residuals receive no inverse-variance weighting. At least 40% of iterations sample the newest keyframe, with remaining iterations drawn from the active submap. Low-opacity Gaussians are pruned midway and after each update.

A projected Gaussian gains one consistency count per processed keyframe when its axial depth agrees with the measured depth within $3\sigma_Z$. After 15 consistent keyframes it becomes stable: its parameters are frozen, and Adam moments are cleared to prevent momentum-driven updates. The threshold translates RTG-SLAM's temporal support criterion to Ariadne's measured keyframe rate and remains an uncalibrated design choice [30]. Growth uses depth seeding; gradient-driven clone/split densification and cross-agent overlap pruning are absent. Per-keyframe affine colour correction is implemented behind a 5000-step delay but was not activated in the reported training experiment.

One live-training interface defect remains: the socket encodes invalid depth as $Z = S = 0$, while the optimizer mask checks finiteness and the upper uncertainty bound without enforcing $Z > 0$. Seeding rejects these pixels, but live GPU optimization could include them in its loss. The synthetic trainer uses NaN invalids and does not exercise this case. Positive-depth validation is required before live GPU-training results can be treated as dependable.

Reconstruction evidence

The synthetic corridor used 60 keyframes, exact ray-cast depth, analytic albedo, and a held-out forward-moving camera. At the live-oriented 15-iteration budget, training-view PSNR rose only from 20.8 to 21.3 dB.

Experiment	Measured result	Evidence boundary
Fused-depth baseline	`ci\_tiny`: 21 mm mean point error, 0.66 completeness within 5 cm, 0.126 m Chamfer; untrained render 17.5 dB PSNR, 0.718 SSIM.	Evaluates the original fused cloud and Gaussian initialization.
Synthetic GPU training	T4, 60 keyframes, two submaps: 100 iterations/keyframe improved median training-view PSNR from 22.2 to 25.1 dB at 715 ms/keyframe; held-out PSNR 17.8–20.1 dB.	One synthetic corridor with exact depth and analytic albedo. The 15-iteration variant gained only 0.5 dB at 123 ms/keyframe.
Live multi-agent service	Three agents, 20 simulated seconds: 71 keyframes, zero tee drops, four submaps, 259,994 Gaussians, 8.3 MB exported; browser rendering passed.	CPU initialization, briefing-derived placements, zero optimizer steps, and no live graph-correction update.

Table 4: Three separately evaluated reconstruction paths. Quality measurements from the fused-cloud and synthetic-training experiments are kept distinct from live integration evidence.

A depth-free diagnostic improved appearance more strongly, but the adopted formulation preserves metric depth supervision.

GPU microbenchmarks on an idle Tesla T4 measured median forward/backward/Adam times of 10.88, 15.33, and 20.72 ms for 100k, 250k, and 500k Gaussians at 640×360. These timings used a simplified loss, smaller synthetic footprints, and exclusive GPU access. They establish a kernel cost envelope; the complete trainer's live throughput remains unmeasured. Unpacked rasterization was faster than packed mode in the evaluated workload.

The live `maze\_demo\_0` run established an uninterrupted path from runtime keyframes to browser-visible Gaussian chunks. Its 259,994 primitives occupied exactly 8,319,808 bytes, and enabling the tee preserved the mission's no-tee state digest. Placement transforms agreed with briefing-derived values to 2.2×10^{-16}. A separate ESDF proximity check placed 92.9% of sampled wall-heavy Gaussian centres within 10 cm of the scene surface, with lower agreement on the floor-heavy submaps. Headless Chromium rendered the resulting scene on an Intel Iris Xe.

These measurements demonstrate streaming, coordinate consistency, and inspectable multi-agent visualization. The accepted live run used pose version zero and CPU seeding throughout. GPU-trained multi-agent reconstruction, independently recovered inter-agent placement, cross-agent Gaussian fusion, and live pose-graph re-anchoring remain outside its demonstrated scope.

Implementation boundary and prior systems

OpenCV 5 supplies the rectified imagery, stereo disparity, and depth-uncertainty inputs. The reconstruction service implements submap anchoring, depth seeding, lifecycle management, training schedules, and atomic chunk publication. Differentiable rasterization runs through gsplat/PyTorch; Spark/WebGL handles interactive browser rendering.

Gaussian-SLAM informs the depth and loss recipe, RTG-SLAM motivates stable/unstable Gaussian management, and LoopSplat provides the closest precedent for rigid submap placement. SplaTAM and Photo-SLAM demonstrate complementary approaches to joint tracking and Gaussian mapping [29, 30, 31, 32, 33]. Their published operating points use different sensors, scenes, and hardware.

Ariadne's demonstrated reconstruction is an inspectable visual layer over validated stereo-inertial geometry. The live service connects three agents to a shared browser scene; the separate GPU experiment establishes measurable refinement on synthetic data. Together they define a functioning reconstruction interface and its current quality and compute limits.

10 OpenCV 5 in the working path

Ariadne's path from camera observations to shared geometry passes repeatedly through OpenCV 5. Image registration, sparse tracking, stereo reconstruction, learned correspondence, robust geometric estimation, and nonlinear optimization all execute through its C++ primitives. Ariadne supplies the stateful machinery around them: landmark management, inertial factors, bundle adjustment, graph consistency, communication policy, volumetric mapping, and mission planning. This boundary makes it possible to examine both what OpenCV

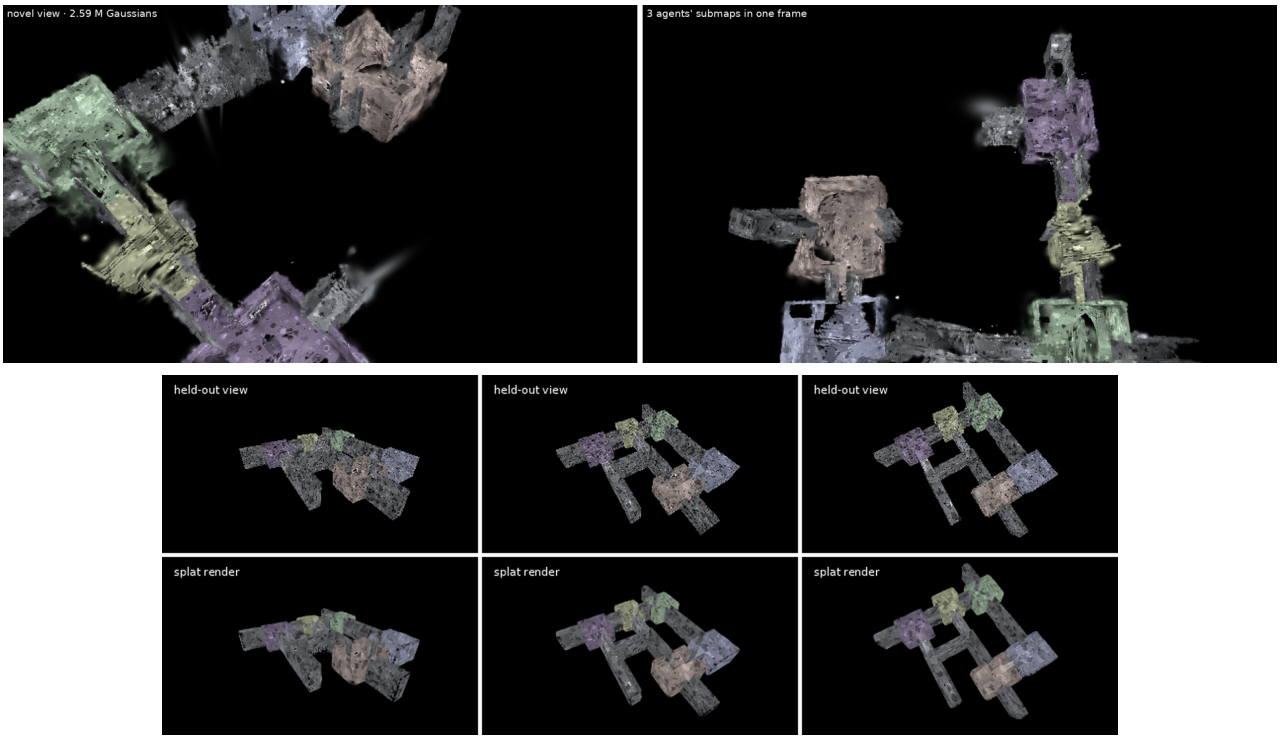


Figure 13: Gaussian-splat reconstruction trained on AWS g4dn (T4). Top: a novel view and the per-agent submaps placed in one frame, from three agents' keyframes of a 200 s mission (training `20261024T121121Z-gpu-ae604c`, render `20261024T132853Z-gpu-1e9c60`; 2.59 M Gaussians; training-view PSNR). Bottom: held-out views of the generated maze next to splat renders (run `20261026T211816Z-gpu-2705f3`, 3,000 steps, 187,405 Gaussians, held-out PSNR 24.36 dB over 6 views, from 23.12 dB at initialisation).

contributes and whether the resulting system improves when a stronger primitive is introduced.

Pinned implementation and numerical contracts

The project builds OpenCV 5.0.0 and matching `opencv\_contrib` sources from pinned commits, with C++17 and Eigen 3.4.1. The contrib build includes `ximgproc`, `xphoto`, `optflow`, and `quality`; build-time probes verify available headers, linked symbols, DNN engines, and stereo interfaces [11]. Run manifests record source revisions, the OpenCV build-information identity, model hashes, thread settings, and configuration. ROS 2 Jazzy and Gazebo Harmonic provide the simulated sensor environment, while the geometry and replay libraries remain usable without ROS.

The pinned release exposed several consequential API details. It includes `cv::ALIKED`, `cv::LightGlueMatcher`, and `cv::ANNIndex`, but has no native `cv::XFeat` class. OpenCV's learned-feature classes also require externally supplied model weights. Its `cv::Volume` interface supports volumetric fusion but does not expose the per-voxel weight and read operations required by Ariadne's planner. These findings were resolved against the installed implementation and retained as explicit algorithm and dependency decisions.

Visual primitives and learned inference

At frame rate, `cv::calcOpticalFlowPyrLK` propagates existing tracks through a three-level pyramid with a 21-pixel window. Forward/backward checks and seeded USAC fundamental-matrix estimation remove inconsistent motion. ORB replenishes features on an 8×6 grid, maintaining a target of 400 active tracks. When landmark-bearing tracks fall below 30, descriptor re-association against healthy local landmarks is followed by seeded PnP, requiring at least 12 inliers before track identities are restored. Stereo depth comes from rectified `StereoSGBM`, with disparity uncertainty propagated into landmark admission and subsequent optimization.

Stereo implementation details affected accuracy directly. Online right-image row refinement reduced the median vertical residual on EuRoC V2_01 from 1.46 to 0.09 pixels. Left/right exposure matching improved correspondence stability under unequal camera response. An optional `ximgproc::DisparityWLSFilter` path computes confidence using right disparity and weighted filtering; preserving the original left matcher avoids a factory-side parameter mutation observed in the pinned API. At 752×480, this path adds approximately

Stage	OpenCV 5 primitives	Implementation and measured evidence
Image preparation	`imgproc`, `photo`, `xphoto`; rectification, remapping, filtering	Stereo rectification, row correction, exposure normalization, flat-field compensation, and optional variance-stabilized denoising. Radiometric stages are separately gated and evaluated for their effect on geometry.
Tracking and stereo	`calcOpticalFlowPyrLK`, `buildOpticalFlowPyramid`, `StereoSGBM`, `DisparityWLSFilter`	Pyramidal LK with forward/backward and epipolar rejection; spatial ORB replenishment; metric disparity and uncertainty. WLS confidence is implemented as an optional path, with unchanged left disparity and an unmeasured confidence-quality AUC.
Learned perception	`ORB`, `ALIKED`, `LightGlueMatcher`, `dnn::Net`, `ANNIndex`	Ariadne-assembled XFeat graph, EigenPlaces retrieval, and native ALIKED–LightGlue verification. Reference-output agreement is tested; exact cosine search is the retrieval default. Learned keyframe matching reaches 0.607 pose AUC at 10°.
Geometric verification	`findFundamentalMat`, `findEssentialMat`, `solvePnPRansac`, `triangulatePoints`	Seeded USAC, reprojection checks, stereo relative-pose estimation, and covariance-aware degeneracy rejection. Verified constraints subsequently enter Ariadne's PCM and robust graph backend.
State estimation	`detail::LevMarqBase`, `solvePnPRefineLM`, `projectPoints`	OpenCV drives nonlinear iterations; Ariadne assembles visual-inertial residuals, analytic SE(3) Jacobians, Schur elimination, covariance, and sparse graph systems. Numerical checks agree with independent PnP and GTSAM oracles.
Volumetric planning	`ptcloud`, `PCA`, `filter2D`, `parallel\_for\_`	OpenCV supplies a fusion oracle, frontier-clustering and support primitives, and parallel scoring. Ariadne implements the production voxel-hash TSDF, incremental 3D ESDF, responder clearance, and route search.
Visualization and routing	`findContours`, `approxPolyDP`; image and depth utilities	Contour geometry supports the experimental visibility graph; validated stereo depths initialize Gaussian submaps. The Gaussian trainer, browser renderer, and D* Lite egress solver are separate implementations.

Table 5: OpenCV 5 interfaces exercised by Ariadne and the corresponding system-level implementations. Numerical agreements, subsystem gains, and optional paths are distinguished by their reported evaluation conditions.

37 ms per stereo pass. It remains off by default because its confidence discrimination has not been measured against the planned reference. The release also treats nonpositive `disp12MaxDiff` as a strict consistency setting, which required explicit parameter validation.

The learned front end illustrates how API-level implementation and trajectory evaluation informed one another. Ariadne assembled XFeat's forward graph directly through `cv::dnn::Net::addLayer` and `connect`, using OpenCV convolution, activation, pooling, resize, and elementwise operations. Batch normalization is folded into convolution weights, while the 8×8 spatial unfold is expressed as a fixed stride-eight convolution. An independent `cv::gemm`-based engine provides an operation-level reference. The runtime graph reproduces upstream feature maps within 8×10^{-6} relative error, all 4096 tested keypoint positions, and descriptor cosine agreement of 1.000000. Both engines use credited upstream weights; model training was outside Ariadne's scope.

That numerical fidelity did not translate into better local odometry. XFeat reduced lost tracks but increased EuRoC trajectory error on six of seven previously passing flights. Keyframe cadence and matcher variants failed to reverse the pattern, so ORB remains the adopted EuRoC VO feature path. Learned inference instead serves recognition events where stronger appearance matching can change the graph [6, 7].

EigenPlaces R50-256 runs through `cv::dnn` with one network instance per thread. Its 256-dimensional descriptors are compared by exact cosine search using an OpenCV matrix product. `cv::ANNIndex` remains a scale and persistence ablation: on 297 EuRoC revisit queries, approximate and exact indices returned identical top-ranked results. EigenPlaces achieved 0.626 **R@1** at 376×240, against 0.444 for mean-bit retrieval and 0.167 for random selection; the larger 752×480 model reached 0.677. The smaller export was retained for its substantially lower inference cost. The evaluated simulator revisits reached 0.705 R@1 at 640×360 [10].

Selected overlaps proceed to `cv::ALIKED` and `cv::LightGlueMatcher`. Ariadne exported ALIKED-n16 and the nine-layer LightGlue model from pinned upstream weights, adapting the ONNX graphs for OpenCV's DNN engine. The export preserves upstream coordinate normalization by passing pre-normalized keypoints through `setPairInfo`; using OpenCV's per-axis normalization would change the model's intended coordinate frame.

Reference tests report keypoint differences below 9.3×10^{-5} pixels, descriptor differences below 4×10^{-6}, and identical match sets.

On 4,815 EuRoC retrieval candidates across 11 flights, ALIKED–LightGlue raised pose AUC at 10° from 0.284 **to** 0.607 against ORB, improving or matching every flight. Geometric average precision over the fixed 676 positive pairs rose from 0.303 to 0.807, and LightGlue contributed another 0.042 AUC over ALIKED with mutual-nearest-neighbour matching. Increasing the keypoint cap from 1024 to 2048 brought no measured pose improvement at approximately 2.8× the compute cost [8, 9].

Four confident false verifications nevertheless retained 210–264 essential-matrix inliers despite rotation errors of 37°–59°. Ariadne consequently keeps stereo PnP, covariance and degeneracy checks, pairwise consistency, and robust graph optimization after learned matching. The front-end score measures correspondence quality; the shared map changes only after the full geometric acceptance path succeeds.

OpenCV optimization and Ariadne's estimator

The stereo-inertial estimator uses OpenCV's `cv::detail::LevMarqBase` as its nonlinear iteration driver. Ariadne implements the backend: manifold residuals, exact left-perturbation SE(3) Jacobians, landmark Schur elimination, sparse normal-equation assembly, gauge fixing, and covariance recovery. OpenCV's driver controls damping and step acceptance; the reduced systems are factored with Eigen. For the windowed visual-inertial problem, the damped normal equations have the block structure

$$\begin{bmatrix} H_{ss} + D_s & H_{sl} \\ H_{ls} & H_{ll} + D_l \end{bmatrix} \begin{bmatrix} \delta s \\ \delta l \end{bmatrix} = - \begin{bmatrix} g_s \\ g_l \end{bmatrix}, \qquad H_{\text{red}} = H_{ss} + D_s - H_{sl}(H_{ll} + D_l)^{-1} H_{ls}, \tag{13}$$

where s contains pose and inertial states and l contains landmarks. Independent landmark blocks are eliminated before solving the reduced state system. The adopted window contains ten keyframes and includes stereo reprojection, preintegrated IMU, and bias random-walk factors.

The split is checked numerically. The pose-graph solver agreed with GTSAM 4.3 on 100/100 seeded synthetic graphs, with worst relative energy difference 1.5×10^{-12}. A one-free-pose bundle-adjustment problem agreed with `cv::solvePnPRefineLM` within 1.3×10^{-9} m and 2.8×10^{-9} rad; analytic Jacobians, Schur solutions, and selected covariance blocks also passed independent checks. The built-in `cv::detail::PoseGraph` optimizer was unsuitable for the pinned configuration, which lacked its required Eigen-enabled build path. Ariadne therefore retains its own SE(3) graph representation and optimizer behind the OpenCV LM interface.

On all 11 EuRoC flights, the complete stereo-inertial estimator achieved mean RMS ATE of 0.120 m, improving to 0.066 **m** with loop closure. The latter fell within the stated published stereo-inertial comparison class on 10/11 flights and matched or improved the published ORB-SLAM3 result on 7/11. These are comparisons with published references under the report's stated alignment protocol; V1_03 remains the principal outlier at 0.288 m.

Beyond the camera front end

OpenCV's geometry and image interfaces continue into planning without carrying its system state. The evaluated `cv::Volume` implementation serves as a fusion comparison; Ariadne's production mapper maintains 5 cm voxel-hash TSDF blocks with explicit observation weights, tri-state occupancy, and incremental 3D ESDF queries. `cv::PCA` assists frontier partitioning, `cv::filter2D` aggregates local support and risk statistics, and `cv::parallel_for\_` distributes independent view scoring. Contour extraction and polygonal approximation support the experimental FAR-style visibility graph. D* Lite, CBBA, geometric risk estimation, and Gaussian submap management are implemented separately.

Camera-model support also mattered outside EuRoC. The Hilti 2023 evaluation introduced equidistant Kannala–Brandt cameras through `cv::fisheye::stereoRectify`, `cv::fisheye::initUndistortRectifyMap`, and `cv::fisheye::undistortP`. Synthetic calibration checks reached 10^{-3}-pixel row agreement and relative depth error below 10^{-4}. The real low-light transfer result remained negative, with 0/100 on all three evaluated Hilti configurations. Correct camera-model handling therefore established a valid measurement pipeline without rescuing the estimator's out-of-domain performance.

Compute profile and reproducibility

Measured inference costs explain the runtime division. On a four-thread `g4dn.xlarge` CPU, median inference was 311.8 ms for EigenPlaces at 640 × 360, 1998.6 ms for ALIKED at 1280 × 720, and 921.0 ms for a warm LightGlue pair. Two ALIKED extractions and one LightGlue match cost roughly five seconds on that CPU

configuration. Continuous LK tracking and event-driven learned matching follow directly from these measurements.

Optimization experiments were held to the same numerical and behavioral criteria as the mapping experiments. A proposed ALIKED graph rewrite reduced eight-thread median inference from 1112.7 to 469.9 ms but changed one verification verdict and exceeded the preregistered descriptor tolerance; it was rejected. Reducing the keypoint cap to 512 lost four accepted pairs, while alternative descriptor wire codings failed their adoption criteria despite substantial byte savings. These results kept the accepted matching configuration stable through the measured communication experiments.

The estimator produced bit-identical trajectories with OpenCV thread pools of 2, 5, and 16 on the tested three-agent replay. Other OpenCV DNN paths, particularly ALIKED, showed thread-dependent output differences and retain an explicit per-thread execution contract. Performance profiling located a substantial live bottleneck in per-agent image ingestion and lens processing. Disabling the unnecessary lens path improved the tested runtime; a subsequent pipelined-ingestion variant changed real-time factor from 0.839 to 0.833, within noise, and remains disabled by default.

A native OpenCV 5.0.0 and contrib build also passed its Arm smoke checks on AWS Graviton4, the COOL stack. In selected 16-thread measurements, ALIKED at 1280×720 required 530 ms on c8g against 829 ms on c7i, a runtime ratio of 0.64; LightGlue and EigenPlaces reached ratios of approximately 0.85–0.88. These compare stock OpenCV builds on different processors.

The final recorded AWS pre-merge gate passed 54/54 integration checks and 708/708 C++ tests using the pinned OpenCV environment. Its render was reused, and the scientific results above retain their individual dataset, seed, revision, and run-manifest provenance. Reproducibility is strongest at these recorded boundaries: MCAP replay, seeded geometric estimation, reference-model agreement, and explicitly versioned build artifacts. Floating-point execution and SIMD selection still require architecture-specific checks.

Across these experiments, OpenCV 5 provides a substantial and testable computational foundation. Its primitives support both classical geometry and learned perception, while measured accuracy, numerical agreement, communication cost, and execution time determine which configurations enter Ariadne's working path.

11 What the experiments establish

Ariadne's experiments follow the path of information through the system: from a visual correspondence to an estimated pose, from local trajectories to a consistent shared map, and from observed geometry to an actionable route. Each stage was evaluated against a matched baseline, an independent numerical oracle, or ground truth. Together, these measurements explain the architecture that emerged from implementation, including several choices that changed when their expected benefits failed to materialize.

What the comparisons reveal

The strongest trajectory result is the effect of loop closure on the same eleven EuRoC flights, reducing mean error by approximately 45%. The final 0.066 m mean compares with ORB-SLAM3's published 0.035 m mean, while improving or matching its reported result on seven individual sequences [12]. The numerical backend was independently checked against GTSAM, OpenCV PnP refinement, and SE-Sync objectives [14]; its agreement confirms the estimation machinery underlying the dataset experiments.

Collaboration produced the largest change in map reliability. A known-entry Mahalanobis gate, followed by PCM and GNC-TLS, suppressed metre-scale false merges to centimetre-scale error. In synthetic outlier campaigns, PCM–GNC handled up to 90% false closures in groups of ten and 70% in two growing clusters, while retaining planted true connections. Spectral exchange achieved near-optimal connectivity utility on the evaluated small instances, supporting the communication-aware selection developed from Tian et al. and the sparse collaboration principles of Swarm-SLAM and Kimera-Multi [17, 19, 20].

The geometric map carried these estimates into responder-scale decisions. Five-centimetre TSDF fusion improved occupied-space precision, hybrid body clearance preserved traversable passages, and all twelve evaluated egress routes agreed with ground-truth passability. Incremental D* Lite reproduced exact shortest-path solutions after randomized map changes [26]. The corresponding multi-agent allocator reached 99.4% of the exact assignment score on its evaluated linked replans, with sparse consensus traffic of 11.8 kB against 123.4 kB for dense exchange [15].

The implementation also closes the operational loop. Live sensing, planning, movement, multi-agent communication, model orchestration, and the browser console share recorded mission state. Gaussian submaps turn the

Subsystem	Measured result	Evaluation
Local matching	ALIKED–LightGlue raised pose AUC@10° from **0.284 to 0.607**; geometric AP rose from 0.303 to 0.807.	4,815 matched EuRoC pairs, 11 flights; ORB baseline.
Place recognition	EigenPlaces reached **0.626 R@1**, against 0.444 for mean-bit and 0.167 for random retrieval. Int8 descriptors preserved top-1 results at 258 B versus 1,024 B.	297 EuRoC queries; exact and approximate index agreement.
Stereo-inertial SLAM	Loop closure reduced mean ATE from **0.120 to 0.066 m**; the final trajectory matched or improved published ORB-SLAM3 results on 7/11 flights.	Same-run EuRoC ablation; published-system reference.
Numerical estimation	Pose-graph optimization agreed with GTSAM on **100/100** graphs, with relative energy error $\leq 1.5 \times 10^{-12}$; seven SE-Sync benchmark graphs received valid certificates.	Independent numerical and optimization oracles.
Robust collaboration	Known-entry gating reduced worst scene-0 seed-mean merged ATE from 3.79 m to **0.014675 m**, retaining all **560** tested true edges.	Paired gated/ungated multi-agent experiments.
Budgeted exchange	Spectral selection reached **0.970** mean utility relative to exact optimum, against 0.936 for greedy selection.	60 enumerated synthetic instances.
Volumetric mapping	TSDF coverage improved from 0.933 to **0.942**; occupied precision rose from 0.707 to **0.865**.	Matched replay against the initial occupancy mapper.
Traversability and risk	Responder traversability achieved **1.000 precision, 0.900 recall**; the structural-risk proxy reached AUC 0.831.	Ground-truth traversability and procedural hazard labels.
Safe egress	All **12/12** planned routes were ground-truth valid; the final 4.44142 m route matched the reference geodesic. D* Lite used 7.6× fewer vertex expansions.	Mission spool and 180 randomized graph updates.
Exploration and allocation	CBBA matched centralized greedy on **40/40** replans, averaging 0.994 of the exact optimum in linked cases. Ariadne attained 0.757 final maze coverage.	Two-agent allocation replay; six-policy exploration sweep.
Active exposure	Recovered from a two-stop overexposure in **2.35 s**, reducing ATE from 19.1 to 16.1 mm against fixed exposure.	Matched degraded-vision replay.
Gaussian reconstruction	Synthetic training-view PSNR improved from **22.2 to 25.1 dB**; the live three-agent service produced 259,994 Gaussians across four submaps.	T4 training experiment; 20 s live reconstruction.
Agentic orchestration	Bedrock completed 100 joint replans over 500 simulated seconds; classical and model-driven runs each reproduced their recorded state in two replays.	Matched two-agent live chooser experiment.
System validation	Estimator trajectories were bit-identical across 2, 5, and 16 OpenCV threads; the AWS integration gate passed **54/54** checks and **708/708** C++ tests.	Seeded MCAP replay and pinned AWS build.

Table 6: Principal experimental results, retaining the comparator and evaluation setting for each claim. Matched ablations, numerical oracles, and published-system references are identified separately.

resulting observations into an inspectable scene, while deterministic replay preserves the geometric and decision traces. The reconstruction trainer's synthetic quality measurements and the live multi-agent streaming results characterize their respective compute paths.

Decisions shaped by experiments

Several promising designs were revised after direct evaluation. XFeat reproduced its upstream network accurately and reduced tracking losses, but increased trajectory error on six of seven passing EuRoC flights; ORB remained the local VO feature. An ALIKED graph optimization achieved 2.37× faster inference but changed geometric-verification decisions, so the verified graph was retained. Lower-bit local descriptors likewise remained experimental after failing the registered adoption criteria.

The next-best-view planner improved coverage per metre and eventual map completion, reaching 0.757 coverage on the maze against 0.713 for coverage-greedy. Its preregistered coverage-time AUC nevertheless ranked third of six policies in both scenes. Execution traces identified repeated turning as the principal cost, motivating heading-aware command planning. A concurrent ingestion optimization intended to raise real-time factor from approximately 0.84 toward 1.0 instead measured 0.833 against 0.839 for its control and was discarded.

Degraded-vision experiments further clarified the operating regime. Active exposure recovered from saturation, while brightness-matched carried illumination remained harder for geometric tracking than fixed lighting. Transfer to the Hilti 2023 low-light sequences failed the established accuracy criterion across all three configurations, and realistic, temporally correlated dust modelling remained outside the completed evaluation. These findings delimit the tested visual conditions and preserve concrete directions for subsequent robustness work.

The resulting system combines classical geometric estimation, selective learned perception, robust multi-agent consensus, volumetric planning, and task-level orchestration. Its principal contribution lies in how these

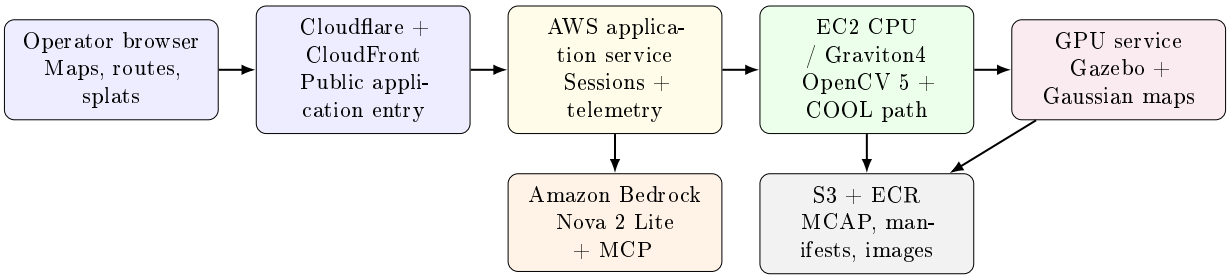


Figure 14: Ariadne's AWS deployment architecture. The browser exposes mission telemetry and reconstruction, CPU services maintain geometry, GPU services process simulation and Gaussian maps, and Bedrock handles bounded task-level decisions. Versioned artifacts connect live execution with reproducible evaluation.

components exchange evidence: candidate observations are priced before transmission, geometric constraints are checked before merging, and responder routes are grounded in observed clearance. The experiments demonstrate that this discipline can preserve map consistency and produce usable routes even when individual visual matches and communication opportunities are unreliable.

12 AWS deployment, accelerated vision, and reproducibility

Ariadne's operator console frontend is hosted on AWS at https://www.ariadne.cynicpixel.xyz/ (Cloudflare DNS, Amazon CloudFront, S3 origin). The hosted page serves the console with a generated scene; the live simulation worker is not attached to it, and a live run is demonstrated by screen share on request. The deployment brings together OpenCV-based estimation, collaborative mapping, Amazon Bedrock task decisions, and Gaussian reconstruction. The same software can consume live simulator streams or recorded MCAP observations, allowing an operator-visible mission to be examined again under identical inputs.

Cloud architecture and operator interface

The deployment separates CPU geometry, GPU rendering, model orchestration, and persistent artifacts. EC2 c7i instances provide high-throughput replay and evaluation; Graviton4 c8g/m8g instances host the arm64 OpenCV execution path; GPU instances support Gazebo and Gaussian reconstruction. Amazon Bedrock Converse invokes Nova 2 Lite for bounded task-level decisions, while S3 and ECR preserve recordings, results, and container images.

The FastAPI operator service exposes session controls, recorded playback, and WebSocket telemetry. Its interface shows agent trajectories, camera feeds, tracked features, communication traffic, map uncertainty, exploration tasks, and responder routes. The three-agent console records camera feature overlays with 491, 371, and 298 points in one demonstrated frame; the collaboration panel exposes verified inter-agent edges and measured bytes, with controls for link interruption, restoration, and false-edge injection. A separate reconstruction pane renders Gaussian submaps through Spark 2.3.1 and Three.js 0.180.0, alongside a distinctly labelled ground-truth scene.

OpenCV 5, COOL, and native Graviton execution

The CPU vision pipeline was structured around OpenCV operations that recur across every agent and frame: rectification and `cv::remap`, image conversion, pyramidal LK, filtering, stereo preparation, resizing, and feature preprocessing. These stable interfaces also define the Graviton acceleration workload, independently of which optional restoration or learned feature variants survive evaluation.

Ariadne pins OpenCV and `opencv\_contrib` to 5.0.0. The corresponding Graviton4 COOL distribution is OpenCV's Cloud Optimized OpenCV 3.1, built on the OpenCV 5.0 line with Arm-specific optimizations. Its published C++ SDK resides under `/opt/cool/cpp\_sdk`; the build configuration can select its CMake package through `OpenCV\_DIR=/opt/cool/cpp\_sdk/lib/cmake/opencv4`. The vendor's CMake directory retains the historical `opencv4` name despite the OpenCV 5 implementation. The same source tree also builds against the project's stock arm64 OpenCV installation.

Reproduction is defined at the library boundary. `infra/docker/core.Dockerfile`, `docker-bake.hcl`, `infra/scripts/core.sh`, and the root CMake configuration establish the pinned build. The COOL preparation path is documented by `infra/cool/day1.sh`, `infra/cool/probe.cpp`, and the `spinup.sh` COOL profile. The build uses `CMAKE\_SKIP\_RPATH=ON`

and checks installed binaries for embedded RPATH/RUNPATH, preserving selection through `LD\_LIBRARY\_PATH`. Each run records the active OpenCV version, CPU features, and an FNV-1a hash of `cv::getBuildInformation()` obtained from the loaded library. This identifies the actual binary implementation used to process a recording.

The native Graviton4 baseline was exercised on c8g with OpenCV 5.0.0 and contrib. At 16 threads, ALIKED inference at 1280×720 measured 530 **ms** on c8g against 829 ms on c7i, or 0.64× the x86 execution time. LightGlue and EigenPlaces measured approximately 0.85–0.88× the corresponding x86 times. These stock-build comparisons establish the arm64 workload baseline; a COOL measurement is produced by the `infra/cool/day1.sh` protocol and identified by its loaded-library manifest. The controlled COOL comparison uses identical inputs, instance type, thread settings, and executable, with the OpenCV implementation as the experimental variable.

Seeded simulation and recorded missions

The evaluation environment is generated on a 6 m room-maze lattice, with seeded corridors, loops, dead ends, entrances, target regions, and responder-scale bottlenecks. Scene identity records the generator version, preset digest, seed, and accepted generation attempt. Gazebo Harmonic renders stereo imagery under configurable illumination, while ROS 2 Jazzy connects cameras, IMU streams, movement, and operator interfaces. Collision-checked kinematic motion provides canonical poses and inertial references.

Sensor streams, motion, calibration, and ground truth are recorded into MCAP. Replay applies seeded image degradation and IMU perturbations to the same observations, permitting matched algorithm comparisons without repeated scene rendering. Run manifests identify source revisions, image digests, OpenCV build fingerprints, model weights, agent count, thread settings, seeds, and planner configuration.

The three-agent, 200-second mission supplied a particularly useful systems stress test. Its recording contains 4,001 estimator poses per agent. A synchronization defect exposed by replay disagreement was corrected, after which two replays reproduced the recorded trajectory state exactly. On c7i.4xlarge, trajectories also remained bit-identical with OpenCV thread pools of 2, 5, and 16. Subsequent profiling located the principal CPU cost in per-agent frame ingestion, guiding the adopted thread-pool and lens-processing configuration.

Bedrock orchestration and live interaction

The live planner submits a joint decision context to Nova 2 Lite through Bedrock Converse. MCP tools expose map renderings, uncertainty, routes, candidate tasks, and agent state. Model responses select planner-generated candidates and supply rationales; the deterministic runtime validates the selections and dispatches commands. The console presents these decisions alongside the observations and route state that informed them.

The first accepted 90-second experiment completed 18 joint replans and 36 per-agent model assignments without a provider error or fallback. A matched classical–Bedrock experiment subsequently held the scene, seed, renderer, runtime, and source revision fixed. Bedrock changed agent 0's selection on 6/18 replans and chose the highest-ranked candidate 31/36 times, against 26/36 for the classical chooser. Both arms passed deterministic replay checks.

The longer matched experiment ran for 500 simulated seconds. Each chooser completed 100 **joint replans and** 200 **commands**, recording 10,001 frames per agent. Two independent replays per arm reproduced the live state digests and each other exactly. The 90-second Bedrock arm measured 5.8 s median and 7.45 s p90 orchestration latency, including tool interactions. These experiments establish sustained multimodal tool orchestration, controlled behavioral differences, and a reproducible decision trace.

GPU reconstruction and artifact delivery

The GPU reconstruction service receives rectified RGB, stereo depth, uncertainty, and anchor-relative poses through a bounded keyframe stream. Its gsplat/PyTorch path supports incremental Gaussian refinement and exports atomic `.splat` chunks with independently versioned placement transforms. The browser renders these chunks using client-side WebGL, allowing incident command to inspect the reconstructed interior from new viewpoints.

In the three-agent integration, 71 keyframes arrived without transport drops and produced four submaps, 259,994 **Gaussians**, and 8.3 MB of browser-ready geometry. A separate Tesla T4 training experiment improved median synthetic training-view PSNR from 22.2 to 25.1 dB over 60 keyframes. The live streaming and GPU-training results retain distinct run identifiers, while sharing the same submap and export contracts.

Terraform manages networking, IAM, compute, ECR, versioned S3 storage, and budget controls. Containers and artifacts are published with digests; private buckets use encryption, TLS-only access, and lifecycle policies.

GitHub OIDC supplies scoped CI credentials, while EC2 jobs use IMDSv2, SSM, and automatic termination. A host-side Spot dispatcher schedules CPU evaluation jobs.

The recorded AWS pre-merge gate passed 54/54 **checks and** 708/708 **C++ tests** in 999 s, at approximately US$0.035 of Spot compute. The measured three-agent replay reached real-time factor 0.836 on c7i.4xlarge. Together, the cloud execution, public entry point, operator interface, model traces, and versioned artifacts provide a reproducible route from simulated observation to collaborative map, selected action, and visual inspection.

13 Limitations and operating envelope

Ariadne's strongest evidence comes from reproducible stereo-inertial estimation, robust map merging, volumetric planning, and multi-agent mission execution. These capabilities have been evaluated through public datasets, seeded simulation, numerical oracles, and recorded cloud runs. Their operating envelope remains shaped by the geometry and sensing assumptions of those experiments.

Perceptual aliasing remains a central challenge when agents have unknown relative starting poses. Known-entry constraints substantially improve map-merging reliability, while PCM can still admit a mutually consistent cluster of false closures. Real-world visual transfer is also demanding: the Hilti low-light evaluation exposed tracking failures during difficult turning sequences, and the adopted next-best-view policy did not meet its coverage-versus-time objective. These results motivate stronger geometric disambiguation and motion-aware exploration.

The simulator captures occlusion, illumination, sensor degradation, and responder-scale obstruction, while leaving particulate dust, material fracture, and evolving structural collapse outside its physical model. Structural risk is consequently a geometric planning proxy. Responder routes are validated against observed clearance in the evaluated environments; field operation would require physical-robot testing, additional environmental sensing, and independent safety validation. These boundaries define the next experimental questions for Ariadne's mapping and decision-support architecture.

14 Responsible use and human authority

Ariadne is intended to support trained responders and incident commanders by presenting observed geometry, traversability, uncertainty, structural-risk indicators, and alternative egress routes. These outputs inform human judgment; a planning score does not authorize entry into a hazardous space. The system requires no personal identification or persistent identity tracking. Its representation concerns the environment: surfaces, observations, agent state, and routes.

Human authority is preserved through explicit approval policies and deterministic planning boundaries. The risk-gate and approval workflow are implemented, although the evaluated live Bedrock missions did not trigger an operator intervention; rejection and fallback dispatch still require end-to-end validation. Unknown space remains excluded from responder traversability, and structural-risk scores remain geometric proxies. Operational rescue use would require physical validation, professional structural assessment, and accountable human command. Ariadne is deployed as a research decision-support system within these boundaries.

15 Reproducibility and evidence trail

Ariadne's repository is both the implementation and the experimental record. Each result is tied to its inputs, executable, configuration, and evaluation procedure, allowing the path from recorded observation to reported measurement to be reconstructed. The central mechanism is record-once simulation: Gazebo renders a mission into MCAP, while estimation, collaboration, planning, and ablation experiments consume the same observations through a ROS-independent replay interface. This separates algorithmic differences from changes in rendering, sensor timing, or simulated motion.

Deterministic input and replay contract

A procedural scene is identified by its generator revision, preset digest, seed, and accepted generation attempt. Its geometry, routes, and kinematic trajectories follow from those inputs. ROS 2 Jazzy and Gazebo Harmonic produce rendered stereo observations; analytically sampled ground-truth poses and IMU measurements are

written into a separate canonical MCAP stream. The two streams are merged by message header timestamp, with topic name breaking ties. Camera observations come from `sensors/`, while inertial and reference poses come from `canonical/`.

This separation addresses a measured recording failure: the live transport lost an IMU sample in a 199-second mission despite delivering the remaining analytic values correctly. Canonical streams instead use the same trajectory sampler as the kinematic mover, with exact sampling grids and zero-initialized CDR padding. The recorder verifies expected topics, stereo pairing, message counts, timestamp intervals, and agreement with an independent trajectory evaluator before accepting a mission. Canonical content is additionally hashed over ordered (topic, timestamp, payload) tuples, independently of MCAP container compression.

Two complementary MCAP interfaces preserve the execution path. The ROS-compatible reader decodes recorded Image, CameraInfo, IMU, PoseStamped, TF, and Clock messages directly through bounded XCDR1 parsers. Internal replay files retain Ariadne's framed wire messages, including their production serialization headers, so replay exercises the same decoding and byte-accounting routines as communication between agents. MCAP uses zstd chunk compression; link traffic is measured separately at the wire serializer.

Sensor degradation is applied during consumption. A partitioned Philox counter-based generator keys random draws by seed, frame, and processing site, making results independent of call ordering and parallel scheduling. Camera noise, exposure transforms, lens effects, and IMU perturbations can consequently be varied over one recording. Project-wide `-ffp-contract=off` prevents compiler-dependent multiply-add contraction from changing the deterministic mathematics. Golden digests cover the deterministic math, Philox streams, degradation operator, and lens transform; tests include adversarial counter ranges and compiler settings to verify that these checks detect unintended changes.

Build identity and reproducible execution

The reference environment pins OpenCV 5.0.0 with matching contrib, Eigen 3.4.1, C++17, ROS 2 Jazzy, and Gazebo Harmonic. `README.md`, the root CMake configuration, `infra/docker/core.Dockerfile`, and `docker-bake.hcl` define the build entry points. For host builds, `ARIADNE_REQUIRE_OPENCV=ON` enforces the complete OpenCV-linked target set; the project CI image provides the corresponding pinned environment.

Each run manifest records the source revision, container image, input and configuration checksums, model hashes, camera calibration, scene identity, random seeds, execution mode, and thread settings. The OpenCV identity includes an FNV-1a hash of `cv::getBuildInformation()` from the loaded library. The arm64 and COOL build paths use explicit CMake package selection and loader configuration; binaries are checked for embedded RPATH/RUNPATH so the selected library can be verified at runtime. This permits stock OpenCV and COOL-compatible Graviton workloads to be reproduced with their respective library identities.

The repository supplies separate entry points for scene validation, mission recording, and cloud evaluation: `worldgen.sh simtest`, `infra/scripts/sim_record.py`, and `infra/eval/premerge_aws_launch.sh`. The latter stages, runs, and tears down the pinned AWS evaluation environment. Each experiment retains its invocation, configuration, manifest, raw measurements, and verdict alongside the resulting figures.

Replay and numerical evidence

Determinism is checked at several boundaries. Identical inputs produced byte-identical internal MCAP recordings within the tested build and architecture; replay preserved framed messages and reproduced the original record-file evaluation gate exactly, including ATE 0.0633155 m, RPE 0.0174459 m, and 8,626 serialized bytes. The three-agent estimator produced bit-identical trajectories with OpenCV thread pools of 2, 5, and 16 on the evaluated x86 host.

Replay also exposed integration defects that ordinary unit tests had missed. During the 200-second three-agent run, a barrier incorrectly waited for two agents, allowing the third agent's keyframes to cross synchronization boundaries. The first divergent trace identified the affected timestamp and keyframe count; correcting the barrier restored identical independent replays. The subsequent live mission and replay checks passed. Similar digest checks verified that optional camera overlays and Gaussian-keyframe streaming preserved the estimator's recorded state.

The matched classical–Bedrock experiments retain the same discipline. The 90-second pair records 18 joint replans per arm; the 500-second pair records 100 replans, 200 commands, and 10,001 frames per agent. Two replays per arm reproduce the live state digests and one another. Model decisions are replayed from recorded responses and tool traces, preserving the evaluated control sequence independently of fresh model inference.

Numerical correctness is tested separately from runtime identity. Pose-graph optimization agrees with GTSAM

Evidence	Repository record
Claims and methodology	docs/results/index.md, docs/decisions.md, docs/README.md
Build and deployment	README.md, infra/docker/core.Dockerfile, infra/cool/, infra/eval/premerge_aws_launch.sh
Simulation and recording	python/dataset_tools/worldgen/, infra/scripts/sim_record.py, python/tests/sim/test_canonical_streams.py
EuRoC SLAM	evaluation/runs/vio/euroc_full_aws/20261022T193510Z-cpux86-655144/
Collaborative merging	evaluation/runs/collaboration/known_entry_gate/verdict/, evaluation/runs/collaboration/byte_table/
Exploration and egress	evaluation/runs/planning/nbv_gate/, evaluation/runs/egress/routes/
Agentic decisions	evaluation/runs/live-agentic/chooser_pair_90s_20261025/, evaluation/runs/live-agentic/chooser_pair_500s_20261025/
Gaussian reconstruction	evaluation/runs/reconstruction/live_e2e_20261024/
AWS integration gate	evaluation/runs/gates/premerge_aws/20261024T083919Z-cpux86-73e3fa/

Table 7: Entry points into Ariadne's evidence trail. Run directories retain the experiment-specific commands, configurations, manifests, metrics, and verdicts; the decision log records the reasoning behind adopted and rejected configurations.

on 100/100 seeded graphs, incremental ESDF and frontier updates agree with full recomputation, and D* Lite matches fresh shortest-path solutions after 180 randomized changes. These independent oracles establish algorithmic consistency alongside the recorded dataset and mission results.

Repository evidence index

The recorded AWS pre-merge gate passed 54/54 integration checks and 708/708 C++ tests in 999 s. Its manifest identifies the tested image, source state, and reused render; individual scientific measurements retain their own run identities. Reproducibility claims are scoped to their verified boundaries: CPU architecture and floating-point code generation can affect exact byte identity, while CUDA optimization and fresh language-model inference have separate execution contracts.

This evidence structure lets a result be traced through the scene and sensor inputs, algorithm configuration, serialized events, numerical outputs, and final evaluation. It also preserves rejected experiments and implementation corrections, so the reported system can be understood through the measurements that shaped it.

16 Conclusion

Ariadne demonstrates how a team of visual agents can turn uncertain observations into shared geometry and a navigable route home. Its stereo-inertial front end, communication-aware map merging, volumetric reconstruction, exploration allocator, and responder-scale egress planner form a connected mapping and decision-support system, with cloud-hosted orchestration, interactive visualization, and reproducible mission records.

The experiments establish measurable gains throughout this pipeline. Across 11 EuRoC flights, loop closure reduced mean ATE from 0.120 to 0.066 m; ALIKED–LightGlue improved pose AUC@10° from 0.284 to 0.607 against ORB. Known-entry verification reduced worst seed-mean collaborative ATE from approximately 3.79 m to 0.0147 m while retaining all 560 tested true edges. CBBA reproduced centralized greedy assignments on 40/40 replans, all 12 evaluated egress routes were ground-truth valid, and the three-agent reconstruction service streamed 259,994 Gaussians into an inspectable browser scene.

The less successful experiments shaped the architecture as well. Perceptual aliasing remains difficult without entry constraints, low-light transfer exposed weaknesses in visual tracking, and next-best-view planning fell short on coverage efficiency over time. These findings sharpen the system's operating assumptions and identify where further robustness is needed.

Ariadne's contribution lies in the continuity between perception, communication, geometric verification, and action. By preserving the evidence behind each map connection and route decision, it offers responders a spatial account of where their agents have been, what they have learned, and which observed passages may

lead back to safety.

References

[1] C. Forster, L. Carlone, F. Dellaert, and D. Scaramuzza. On-Manifold Preintegration for Real-Time Visual-Inertial Odometry. *IEEE Transactions on Robotics*, 33(1), 2017.

[2] V. Usenko, N. Demmel, D. Schubert, J. Stückler, and D. Cremers. Visual-Inertial Mapping with Non-Linear Factor Recovery. *IEEE Robotics and Automation Letters*, 5(2), 2020.

[3] A. Rosinol, M. Abate, Y. Chang, and L. Carlone. Kimera: An Open-Source Library for Real-Time Metric-Semantic Localization and Mapping. In *Proceedings of the IEEE International Conference on Robotics and Automation*, 2020.

[4] F. Dellaert. *Factor Graphs and GTSAM: A Hands-on Introduction*. Georgia Institute of Technology, 2012. https://gtsam.org/tutorials/

[5] J.-Y. Bouguet. *Pyramidal Implementation of the Lucas–Kanade Feature Tracker: Description of the Algorithm*. Intel Corporation, 2001. https://www.cs.ucf.edu/courses/cap4453/bouguetopticalflow.pdf.

[6] E. Rublee, V. Rabaud, K. Konolige, and G. Bradski. ORB: An Efficient Alternative to SIFT or SURF. In *Proceedings of the IEEE International Conference on Computer Vision*, 2011.

[7] G. Potje, F. Cadar, A. Araujo, R. Martins, and E. R. Nascimento. XFeat: Accelerated Features for Lightweight Image Matching. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.

[8] X. Zhao, X. Wu, W. Chen, P. C. Y. Chen, Q. Xu, and Z. Li. ALIKED: A Lighter Keypoint and Descriptor Extraction Network via Deformable Transformation. *arXiv preprint arXiv:2304.03608*, 2023.

[9] P. Lindenberger, P.-E. Sarlin, and M. Pollefeys. LightGlue: Local Feature Matching at Light Speed. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023.

[10] G. Berton, G. Trivigno, B. Caputo, and C. Masone. EigenPlaces: Training Viewpoint Robust Models for Visual Place Recognition. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023.

[11] OpenCV contributors. *OpenCV 5.0 documentation and release notes*. https://docs.opencv.org/5.0/ and https://github.com/opencv/opencv/wiki/OpenCV-5.

[12] C. Campos, R. Elvira, J. J. Gómez Rodríguez, J. M. M. Montiel, and J. D. Tardós. ORB-SLAM3: An Accurate Open-Source Library for Visual, Visual-Inertial, and Multi-Map SLAM. *IEEE Transactions on Robotics*, 37(6), 2021.

[13] R. Mur-Artal and J. D. Tardós. ORB-SLAM2: An Open-Source SLAM System for Monocular, Stereo, and RGB-D Cameras. *IEEE Transactions on Robotics*, 33(5), 2017. https://arxiv.org/abs/1610.06475.

[14] L. Carlone, D. M. Rosen, G. Calafiore, J. J. Leonard, and F. Dellaert. Lagrangian duality in 3D SLAM: Verification techniques and optimal solutions. *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2015.

[15] Y. Choi, L. Brunet, and J. P. How. Consensus-Based Decentralized Auctions for Robust Task Allocation. *IEEE Transactions on Robotics*, 25(4), 2009.

[16] M. Burri et al. The EuRoC micro aerial vehicle datasets. *International Journal of Robotics Research*, 35(10), 2016.

[17] Y. Tian, K. Khosoussi, M. Giamou, J. P. How, and J. Kelly. Near-Optimal Budgeted Data Exchange for Distributed Loop Closure Detection. In *Robotics: Science and Systems*, 2018. https://www.roboticsproceedings.org/rss14/p71.pdf.

[18] J. G. Mangelson, D. Dominic, R. M. Eustice, and R. Vasudevan. Pairwise Consistent Measurement Set Maximization for Robust Multi-Robot Map Merging. In *IEEE International Conference on Robotics and Automation*, 2018. https://robots.engin.umich.edu/publications/jmangelson-2018a.pdf.

[19] P.-Y. Lajoie and G. Beltrame. Swarm-SLAM: Sparse Decentralized Collaborative Simultaneous Localization and Mapping Framework for Multi-Robot Systems. *IEEE Robotics and Automation Letters*, 2024. https://arxiv.org/abs/2301.06230.

[20] Y. Tian, Y. Chang, F. Herrera Arias, C. Nieto-Granda, J. P. How, and L. Carlone. Kimera-Multi: Robust, Distributed, Dense Metric-Semantic SLAM for Multi-Robot Systems. *IEEE Transactions on Robotics*, 2022. https://arxiv.org/abs/2106.14386.

[21] H. Yang, P. Antonante, V. Tzoumas, and L. Carlone. Graduated Non-Convexity for Robust Spatial Perception: From Non-Minimal Solvers to Global Outlier Rejection. *IEEE Robotics and Automation Letters*, 5(2), 2020. https://arxiv.org/abs/1909.08605.

[22] A. Hornung, K. M. Wurm, M. Bennewitz, C. Stachniss, and W. Burgard. OctoMap: An Efficient Probabilistic 3D Mapping Framework Based on Octrees. *Autonomous Robots*, 34, 2013.

[23] H. Oleynikova, Z. Taylor, M. Fehr, R. Siegwart, and J. Nieto. Voxblox: Incremental 3D Euclidean Signed Distance Fields for On-Board MAV Planning. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2017.

[24] Y. Zhou, C. Zheng, Y. Liu, F. Wang, and S. Shen. FUEL: Fast UAV Exploration Using Incremental Frontier Structure and Hierarchical Planning. *IEEE Robotics and Automation Letters*, 6(2), 2021.

[25] C. Cao, H. Zhu, H. Choset, and J. Zhang. TARE: A Hierarchical Framework for Efficiently Exploring Complex 3D Environments. In *Robotics: Science and Systems*, 2021.

[26] S. Koenig and M. Likhachev. D* Lite. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2002.

[27] F. Yang, C. Cao, H. Zhu, J. Oh, and J. Zhang. FAR Planner: Fast, Attemptable Route Planner Using Dynamic Visibility Update. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2022.

[28] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis. 3D Gaussian Splatting for Real-Time Radiance Field Rendering. *ACM Transactions on Graphics*, 42(4), 2023. https://repo-sam.inria.fr/fungraph/3d-gaussian-splatting/.

[29] V. Yugay, Y. Li, T. Gevers, and M. R. Oswald. Gaussian-SLAM: Photo-realistic Dense SLAM with Gaussian Splatting. *arXiv preprint arXiv:2312.10070*, 2023. https://arxiv.org/abs/2312.10070.

[30] Z. Peng, T. Shao, Y. Liu, J. Zhou, Y. Yang, J. Wang, and K. Zhou. RTG-SLAM: Real-time 3D Reconstruction at Scale Using Gaussian Splatting. In *ACM SIGGRAPH Conference Papers*, 2024. https://arxiv.org/abs/2404.19706.

[31] L. Zhu, Y. Li, E. Sandström, S. Huang, K. Schindler, and I. Armeni. LoopSplat: Loop Closure by Registering 3D Gaussian Splats. *arXiv preprint arXiv:2408.10154*, 2024. https://arxiv.org/abs/2408.10154.

[32] N. Keetha, J. Karhade, K. M. Jatavallabhula, G. Yang, S. Scherer, D. Ramanan, and J. Luiten. SplaTAM: Splat, Track & Map 3D Gaussians for Dense RGB-D SLAM. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024. https://arxiv.org/abs/2312.02126.

[33] H. Huang, L. Li, H. Cheng, and S.-K. Yeung. Photo-SLAM: Real-time Simultaneous Localization and Photorealistic Mapping for Monocular, Stereo, and RGB-D Cameras. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024. https://arxiv.org/abs/2311.16728.

[34] B. Yu, H. Kasaei, and M. Cao. L3MVN: Leveraging Large Language Models for Visual Target Navigation. *arXiv preprint arXiv:2304.05501*, 2023. https://arxiv.org/abs/2304.05501.

[35] D. Shah, M. Equi, B. Osinski, F. Xia, B. Ichter, and S. Levine. Navigation with Large Language Models: Semantic Guesswork as a Heuristic for Planning. *arXiv preprint arXiv:2310.10103*, 2023. https://arxiv.org/abs/2310.10103.

[36] A. Rajvanshi, K. Sikka, X. Lin, B. Lee, H.-P. Chiu, and A. Velasquez. SayNav: Grounding Large Language Models for Dynamic Planning to Navigation in New Environments. *arXiv preprint arXiv:2309.04077*, 2023. https://arxiv.org/abs/2309.04077.

[37] B. Yu, H. Kasaei, and M. Cao. Co-NavGPT: Multi-Robot Cooperative Visual Semantic Navigation using Large Language Models. *arXiv preprint arXiv:2310.07937*, 2023. https://arxiv.org/abs/2310.07937.

[38] OpenCV. *OpenCV AI Competition 2026, powered by AWS*. https://opencv.org/opencv-ai-competition-2026.